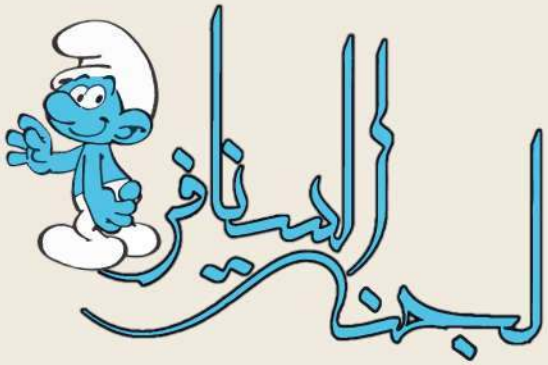
خدمتكم طريق خضناه لرضى الله

2021

أسئلة سنوات ميد

مختبر كيمياء





سنافر البوليتكنك

بسم الله الرحمن الرحيم

نقدم لكم نحن أسرة فريق

(لجنة السنافر)

مجموعة أسئلة اختبارات إلكترونية تم تجميعها خلال الفصول الماضية
سائلين المولى أن يوفقنا وإياكم لكل خير

تنويه

يوجد بعض الأسئلة عليها إجابات قد تحتمل الصواب وقد تحتمل الخطأ

فإن أصبنا فما هو إلا توفيق من الله

وإن أخطأنا فمن أنفسنا

#خدمتكم_طريق_خضناه_لرضى_الله

#الإتجاه_الاسلامي

#بسواعدنا_نبنيها

#لجنة_السنافر

#هي_لله

Burette is used in density of solid Experiment .



Select one:

- True
- False

Graduated cylinder is used in density of solid Experiment .



Select one:

- True
- False

color appear after add the indicator in vinegar analysis is

Select one:

- a. yellow
- b. deep blue
- c. pinke
- d. colorless



Iodine (I_2) will give a purples color when mixing with Water by the means of dissolving.

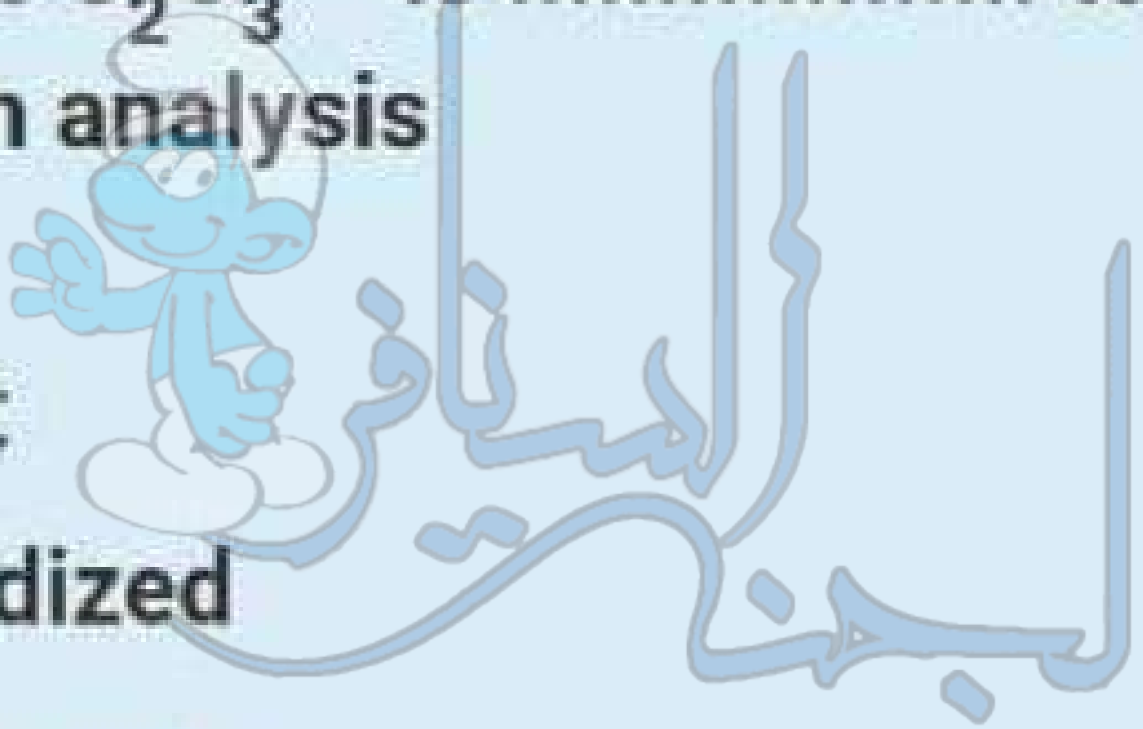
Select one:

- True
- False

Thiosulfate $S_2O_3^{2-}$ is ..... to $S_4O_6^{2-}$ in bleach analysis

Select one:

- A. **Oxidized**
- B. **Reduced**



To avoid chemical loss, urused chemicals should be returned to the stock bottles.



اخترأحد الخيارات

- صح
- خطأ

Which one of the following is correct Concerning equipment used in empirical formula determination Experiment ?

Select one:

- A. **Clay triangle, Bunsen burner, crucible and cover, ring and stand**
- B. **Flask, Bunsen burner, water bath, ring and stand**
- C. **Beaker, Bunsen burner, crucible and cover, Desiccator**
- D. **Clay triangle, , crucible and cover, ring stand, Thermometer**

What is happened when we adding water droplets to the crucible that contain Mg after the first heating is finished ?

Select one:

- A. **NH_3 ammonia gas is evaporated**
- B. **H_2 gas evolving**
- C. **N_2 gas evolving**
- D. **It convert nitride in to H_2O and Oxide**

During an *endothermic* chemical reaction,

Select one:

- A. **The System becomes warmer, and the surrounding undergoes a decrease in temperature.**
- B. **The System becomes warmer, and the surrounding undergoes an increase in temperature.**
- C. **The System becomes cooler, and the surrounding undergoes an increase in temperature.**
- D. **The System becomes cooler, and the surrounding undergoes a decrease in temperature.**

Given the following thermochemical equations:

$$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(L)}$$
$$\Delta H = -56.5 \text{ kJ\mol}$$

$$NH_{3(aq)} + HCl_{(aq)} \rightarrow NH_4Cl_{(aq)}$$
$$\Delta H = 50.5 \text{ kJ\mol}$$

Calculate ΔH for the following reaction (in kJ):

$$NH_4Cl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(L)} + NH_{3(aq)}$$

Select one:

- A. **-9.5**
- B. **+3.5**
- C. **-5.5**
- D. **-107**

A glass funnel is used for filling a burette with a titrant.

Select one:

- True
- False



For the reaction $A+B \rightarrow C+D$. If the order of reactant A and B is 2 and 3 respectively ;which unit represent the rate law constant K unit ?

Select one:

- A. $mol^{-4}\ L^{4}\ S^{-1}$
- B. $mol^{-1}\ L^{2}\ S^{-1}$
- C. $mol^{-1}\ L^{3}\ S^{-1}$
- D. S^{-1}

If the reaction's rate constant k at 25°C is $33\ M^{-1}s^{-1}$ and $45\ M^{-1}s^{-1}$ at 402°C, Calculate the activation energy of the reaction in J/mol .

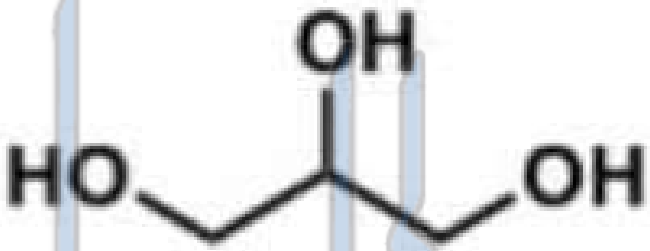
knowing that R=8.3145 J.K/mol and $\ln(k_2/k_1) = -E_a/R\ ((1/T_2)-(1/T_1))$

Select one:

- A. E_a =21.345 J/mol
- B. E_a = 1375.8 J/mol
- C. E_a = 13.758 J/mol
- D. E_a =2134.5 J/mol

1] 

2] 

3] 

Compound number (3) has the highest boiling point

Select one:

- True
- False

Increasing the pressure will increase the solubility of gases.

Select one:

- True
- False

Which one of the following is incorrect concerning kinetics of reaction?

Select one:

- A. **Rate of reaction is measured by the appearance of product with respect time**
- B. **For the reaction which are mth order in A and nth order in B; its overall order equal m + n.**
- C. **Rate of reaction is measured by the appearance of reactant with respect to time**
- D. **In chemical kinetic experiment the rate of reaction measured by the rate of appearance of blue color.**

The equipment in the figure which called desiccator is the suitable equipment that you used to keep the hydrated salt cooled in after heating

Select one:

- True
- False



If the reaction's rate constant k at 25°C is 33 $M^{-1}s^{-1}$ and 45 $M^{-1}s^{-1}$ at 402°C, Calculate the activation energy of the reaction in J/mol .

knowing that R=8.3145 J.K/mol and $\ln(k_2/k_1) = -E_a/R\ ((1/T_2)-(1/T_1))$

Select one:

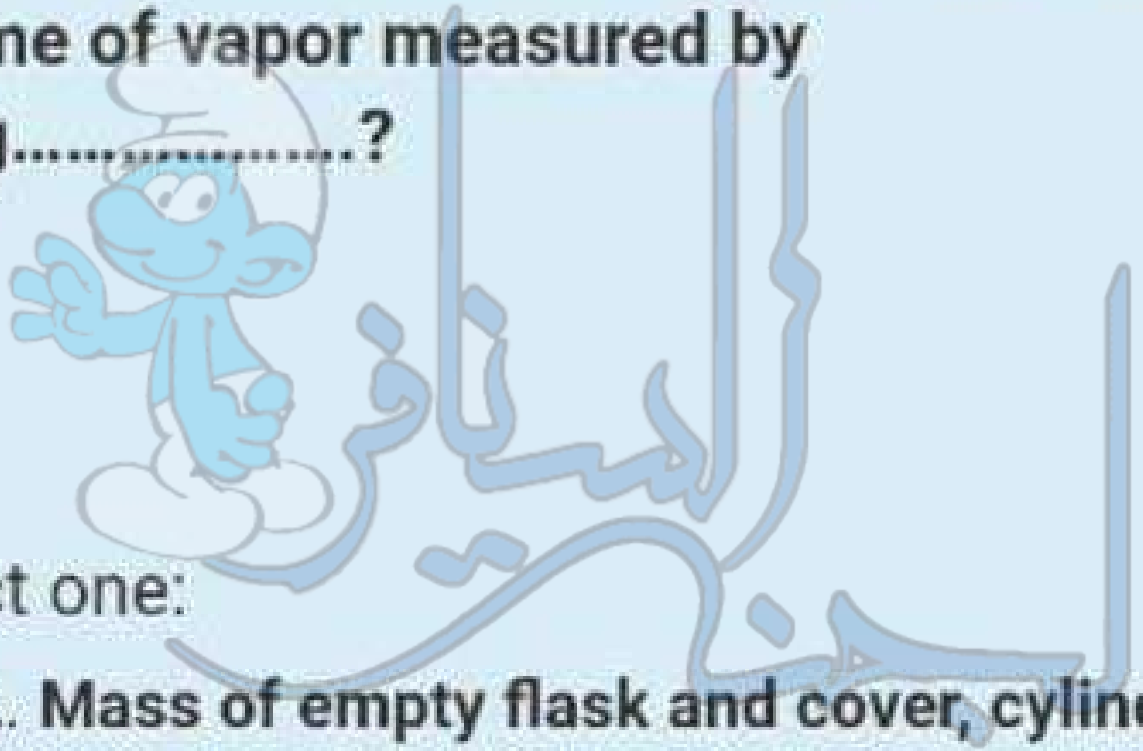
- A. E_a =21.345 J/mol
- B. E_a = 13.758 J/mol
- C. E_a = 1375.8 J/mol
- D. E_a =2134.5 J/mol

Why it is not correct to rinse the burette only with distilled water rather than the solution to be filled in it?

Select one:

- A. **The water will Increase the volume so the concentration will decrease**
- B. **The water will decrease the volume so the concentration will increase**
- C. **The water will Increase the volume so the concentration will increase**
- D. **The water will decrease the volume so the concentration will decrease**

During the molecular weight of volatile liquid experiment. In the end of experiment we record the .....and then the volume of vapor measured by using.....?



Select one:

- A. **Mass of empty flask and cover, cylinder**
- B. **Mass of flask/cover/vapor, flask**
- C. **Mass of empty flask and cover, flask**
- D. **Mass of flask/cover/vapor, cylinder**

Given the following thermochemical equations:

$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(L)}$ $\Delta H=$ -60.0 kJ\mol

$NH_{3(aq)} + HCl_{(aq)} \rightarrow NH_4Cl_{(aq)}$ $\Delta H=$ -50.5 kJ\mol

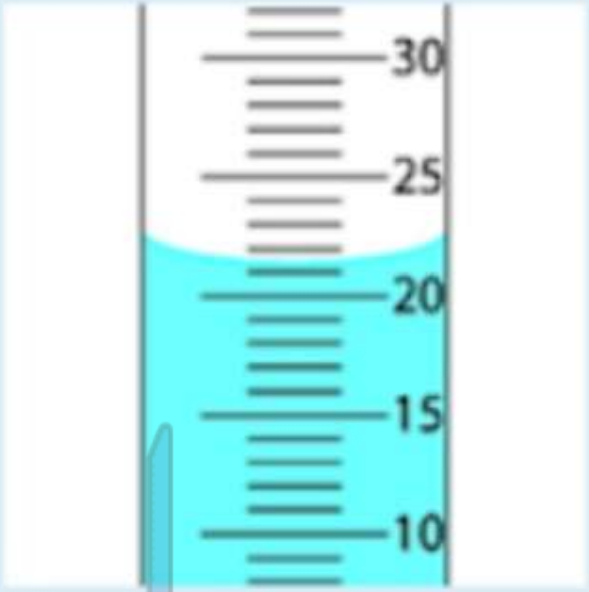
Calculate ΔH for the following reaction (in kJ):

$NH_4Cl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(L)} + NH_{3(aq)}$

Select one:

- A. -107
- B. -9.5
- C. +3.5
- D. -5.5

?What is the volume inside the following cylinder



البحر الصافي

اختر أحد الخيارات

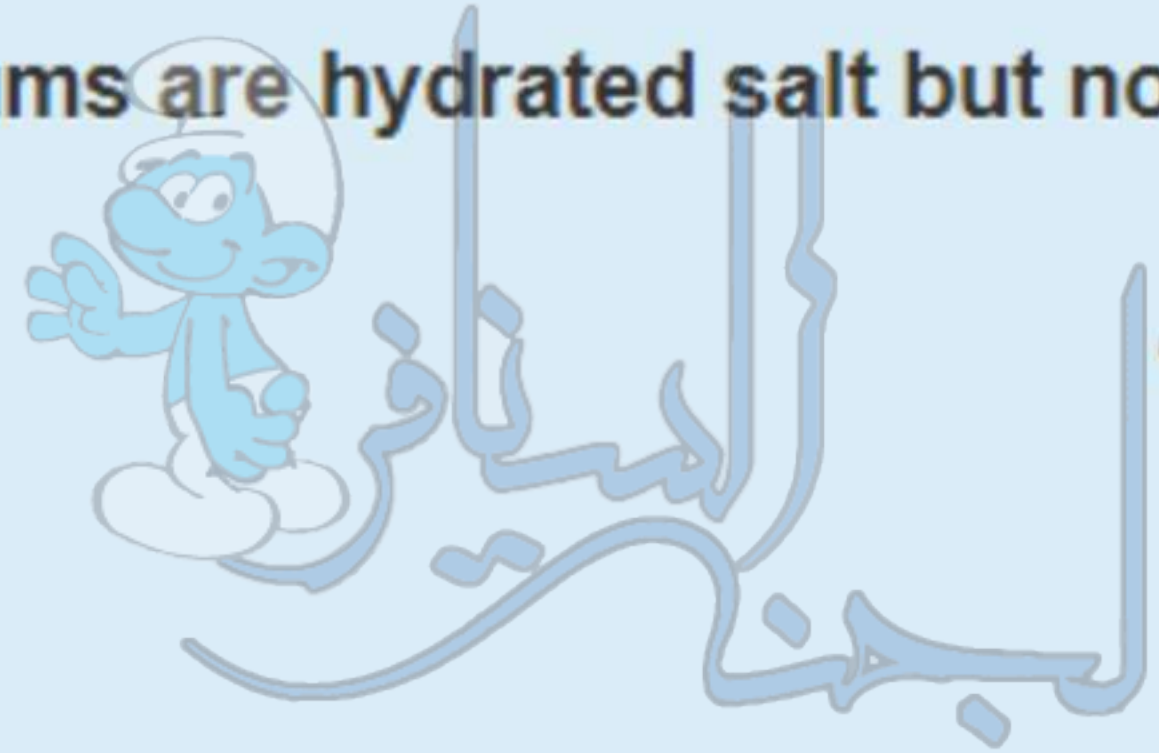
A. 21.5 mL

B. 22.5mL

C. 22.0mL

D. 20.5mL

.All Alums are hydrated salt but not vise versa



اختر أحد الخيارات

- (●) صح
- ( ) خطأ

?? Which of the following pressure will give you the lowest boiling point of benzene

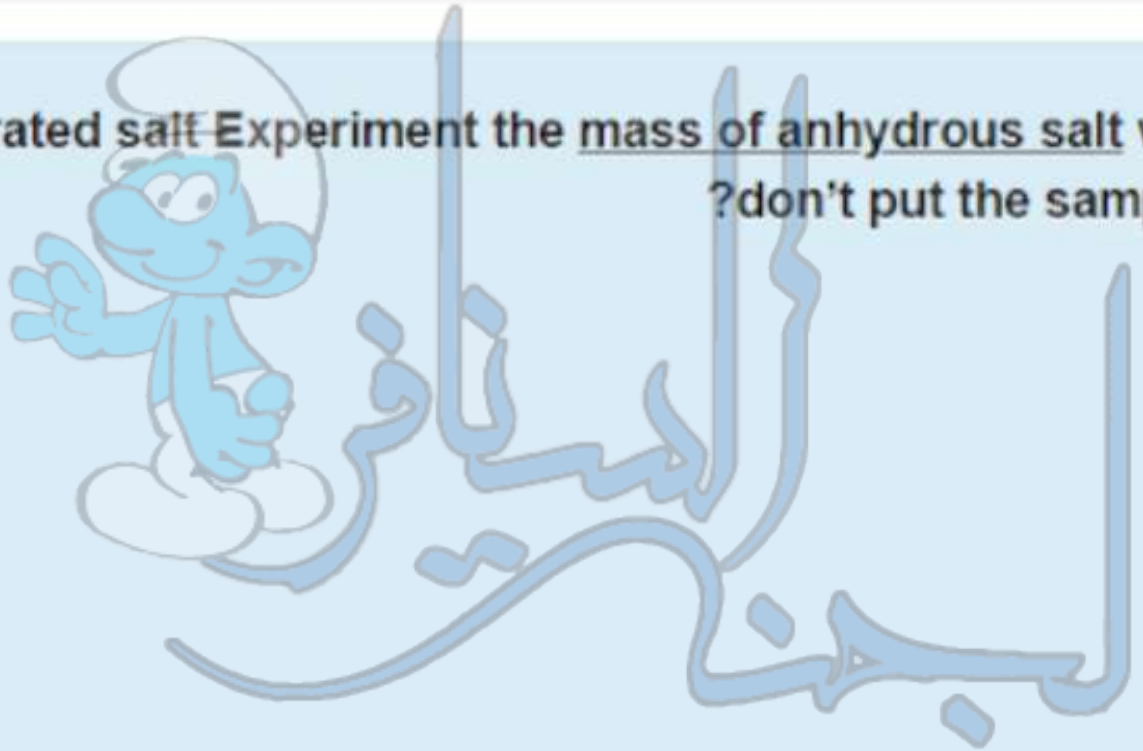
A. 0.

B. 2.5 atm

C. 1 atm

D. 1.95 atm

During the hydrated salt Experiment the mass of anhydrous salt will be ----------if we don't put the sample in desiccator to cool?



اختر أحد الخيارات

A. Increase

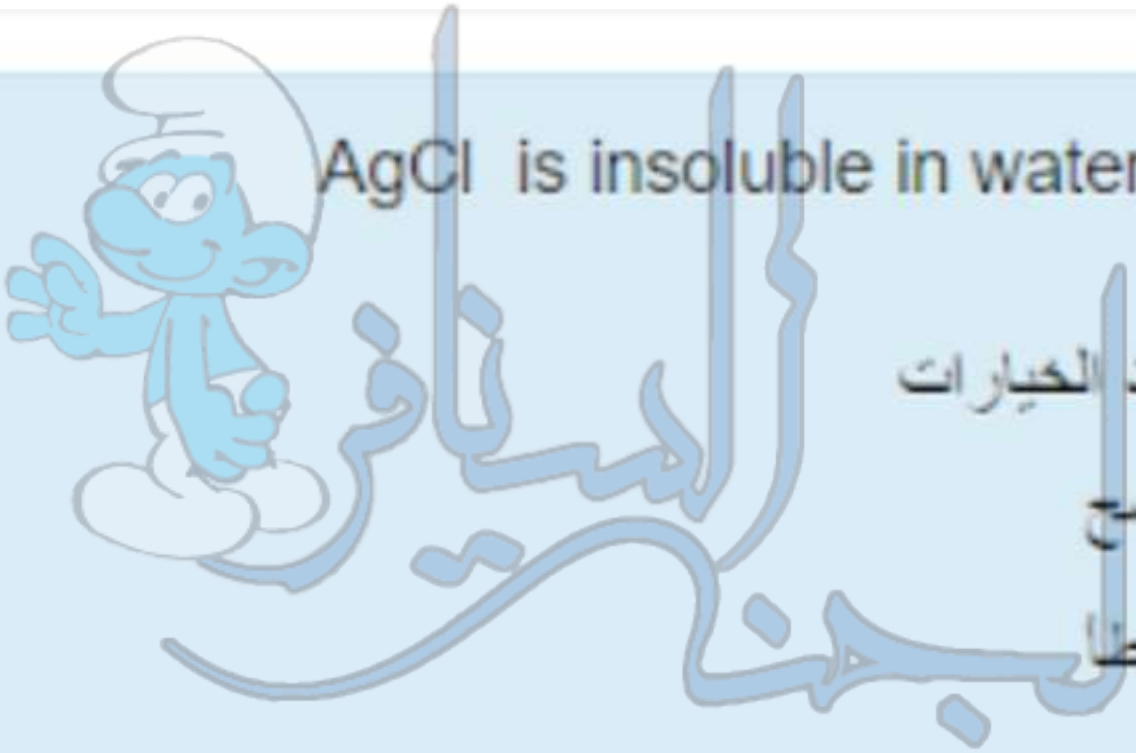
B. not effected

C. Decrease

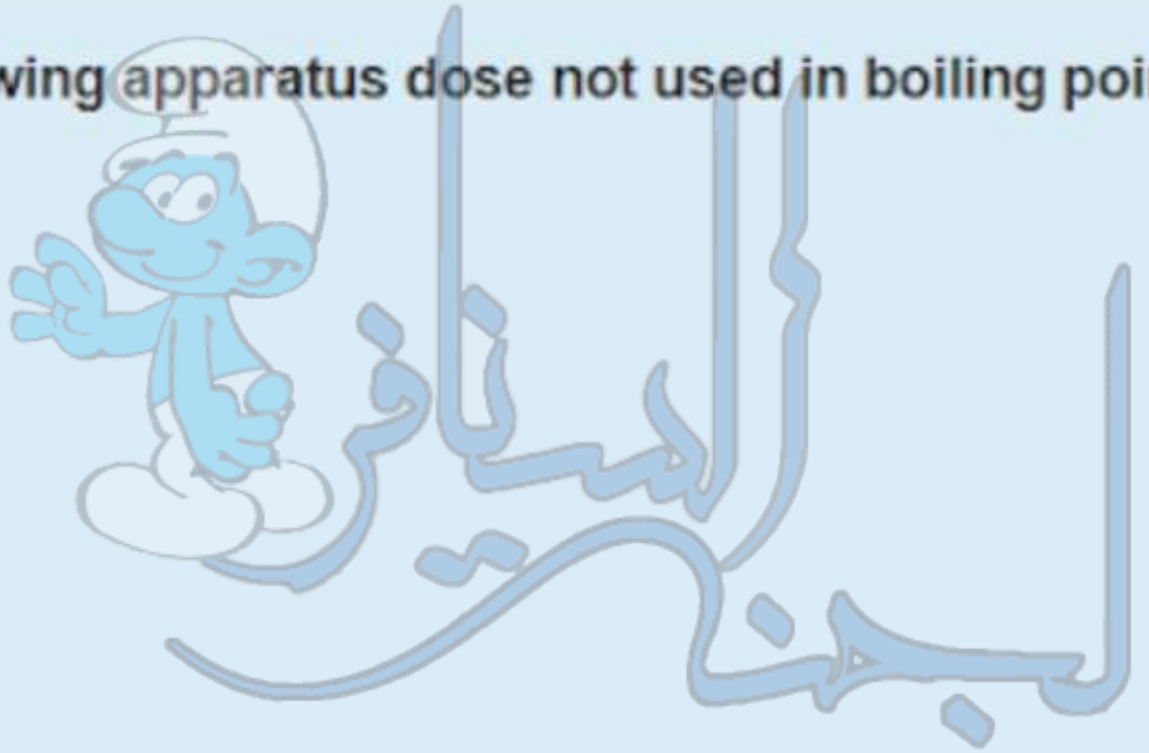
AgCl is insoluble in water H_2O

اختر أحد الخيارات

- صح
- خطأ



Which of the following apparatus dose not used in boiling point determination set up



حدد أحد الخيارات

A. Stand with clamp

B. Thermometer

C. Capillary Tube

D. Flask

?Which of the following compounds have the Lower boiling point



اختر أحد الخيارات

A. $CH_3CH_2CH_2CO_2H$

B. $CH_3(CH_2)_7CO_2H$

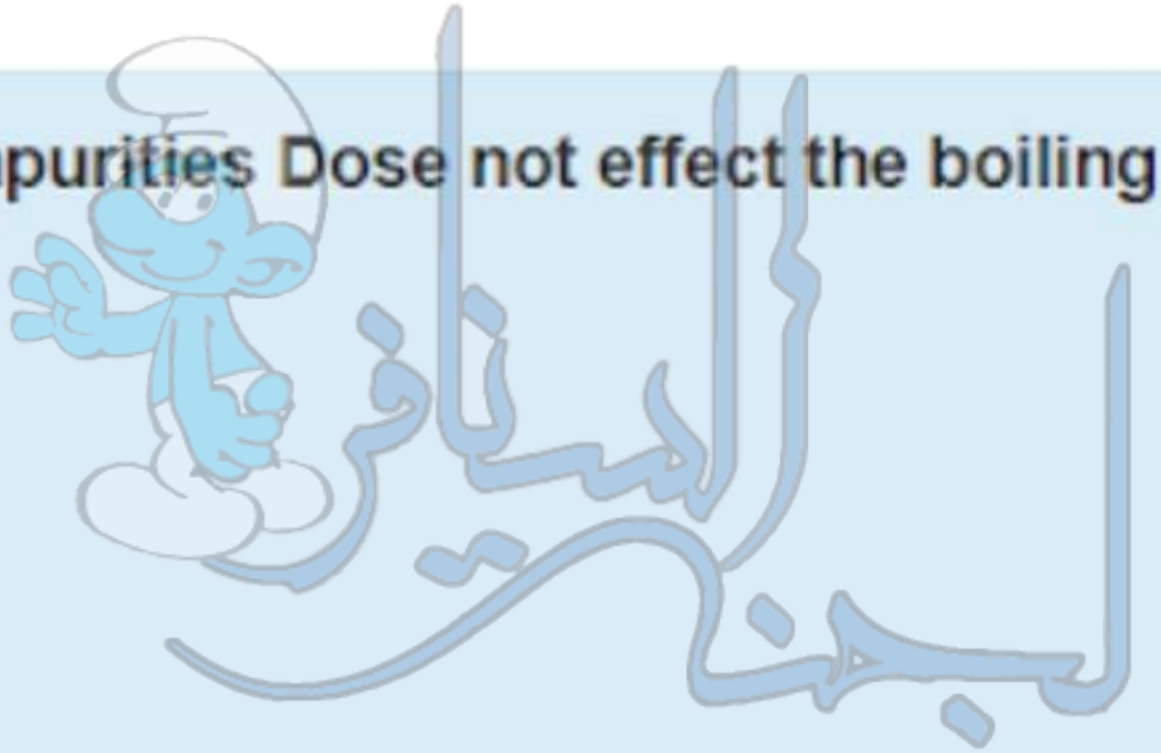
C. $CH_3CH_2CH_2CH_2CH_2CH_2CO_2H$

D. $CH_3(CH_2)_6CO_2H$

insoluble impurities Dose not effect the boiling point of liquid

اختر أحد الخيارات

- صح
- خطا



??Which one of these salts is an Alum



اختر أحد الخيارات

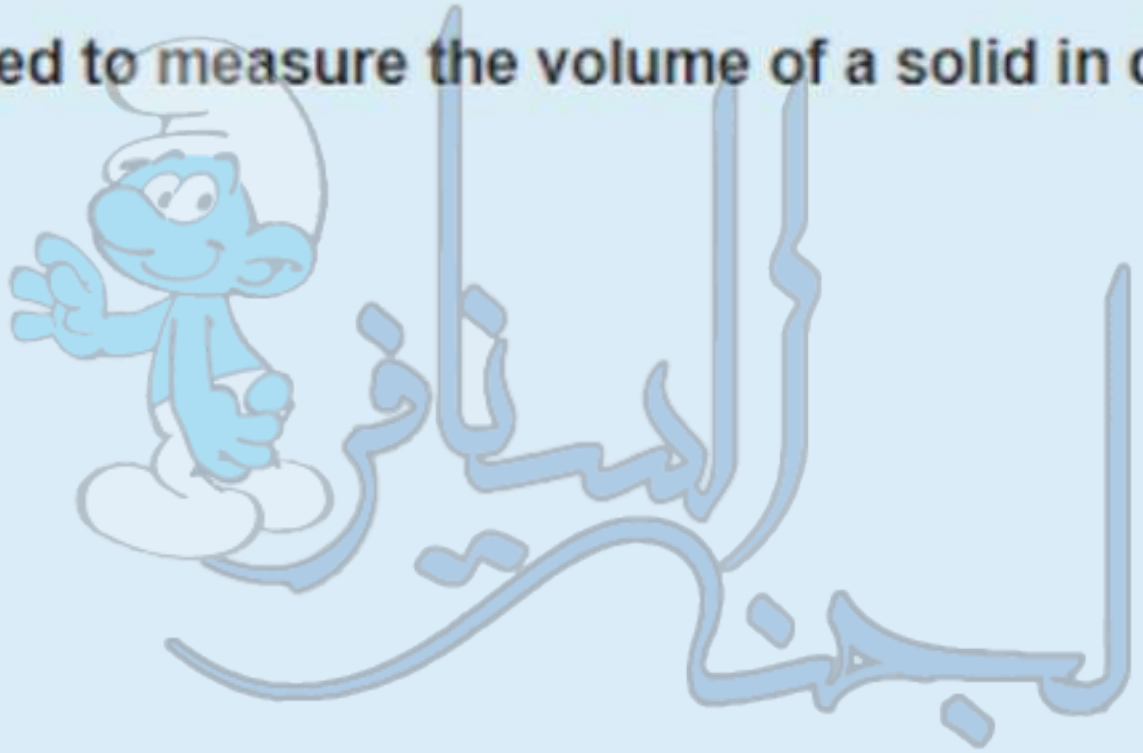
A. $CuSO_4.5H_2O$

B. $KAl(SO_4)_2.12H_2O$

C. $NaAl(SO_3)_2.12H_2O$

D. $CaMg(SO_4)_2.10H_2O$

equipment is used to measure the volume of a solid in density determination
?experimen



ختر أحد الخيارات

A. Thermometer

B. Balance

C. flask

D. Graduated cylinder

:The incorrect statement of the following is

ختر أحد الخيارات

A. You have to tell your instructor when the mercury thermometer is broken.

B. A 10mL graduated cylinder is more precise than a 10mL pipet.

C. Lab coat should be worn all the lab time.

D. Fume hood is the suitable equipment to resist toxic vapor.

Which equipment is used to measure the mass of a liquid in density experiment?

Select one:

- A. Capillary tube
- B. Thermometer
- C. Test tube
- D. Balance

Which equipment is used to measure the mass of a liquid in density experiment?



اخترأحد الخيارات

A. Test tube

B. Balance

C. Thermometer

D. Capillary tube

insoluble impurities Dose not effect the boiling point of liquid



اخترأحد الخيارات

- صح
- خطأ

If the % mass of H_2O = 37.5%, and the molar mass (Mm) of anhydrous salt = 379 g/mol. The value of water of crystallization (X) is: (Mm H_2O =18g/mol).



Select one:

- A. 12
- B. 10
- C. 16
- D. 13

Which of the following apparatus is used in boiling point determination set up to indicate about boiling ?



اخترأحد الخيارات

A. Crucible

B. Stand with clamp

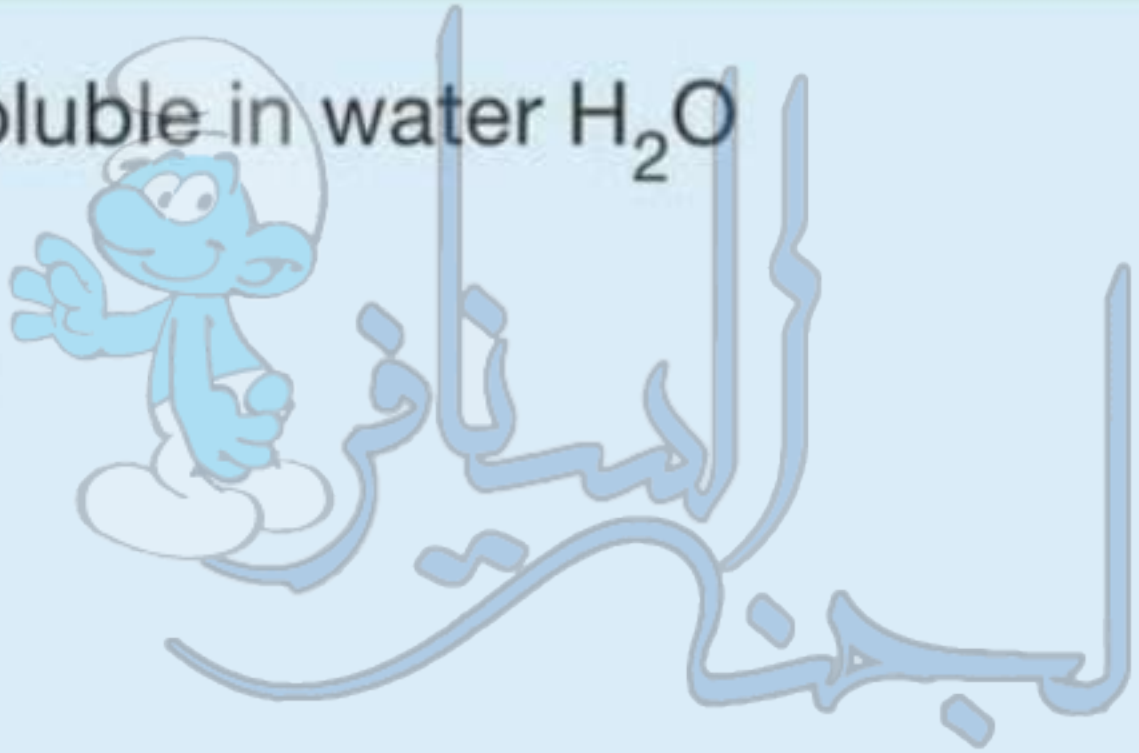
C. Capillary Tube

D. Thermometer

PbI_2 is insoluble in water H_2O

Select one:

- True
- False



Which of the following pressure will give you the highest boiling point of oil ?



اخترأحد الخيارات

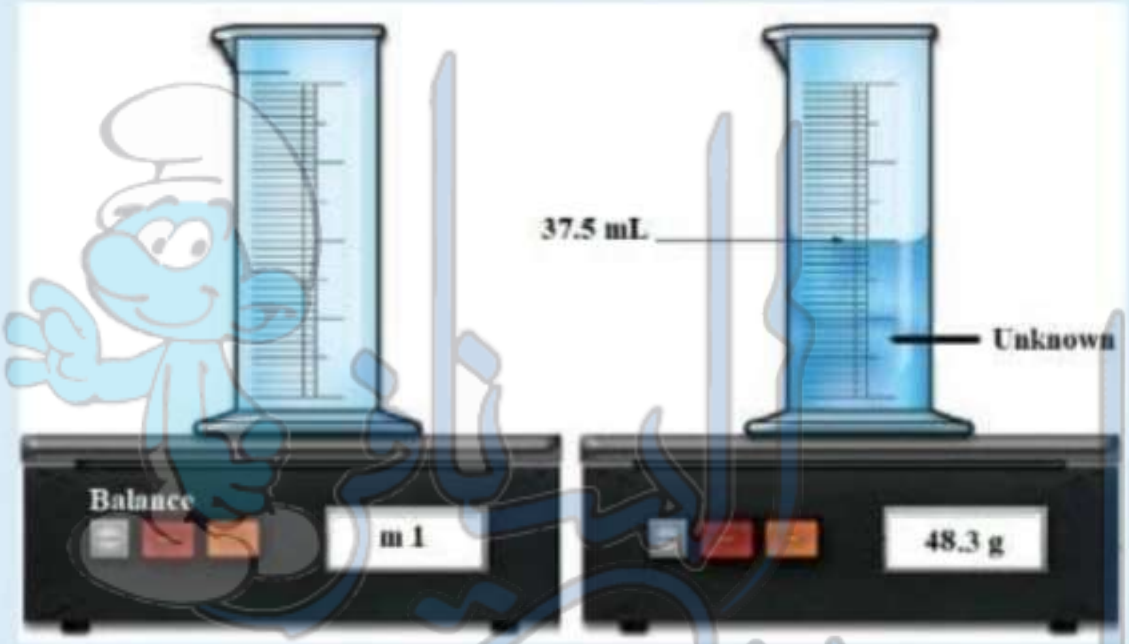
A. 2.5 atm

B. 5.5 atm

C. 0.52 atm

D. 1 atm

According to the photo beside ,if the density of the unknown liquid = 0.87 g/mL, then the mass (m1) equal (in grams) :



Select one:

- A. 80.9 g
- B. 37.5 g
- C. 15.7 g
- D. 32.6 g

For most of the gases solutes, What happens to the solubility if the pressure decreases?



Select one:

- A. Solubility increases
- B. Solubility decreases
- C. Solubility stays the same
- D. None of the choices

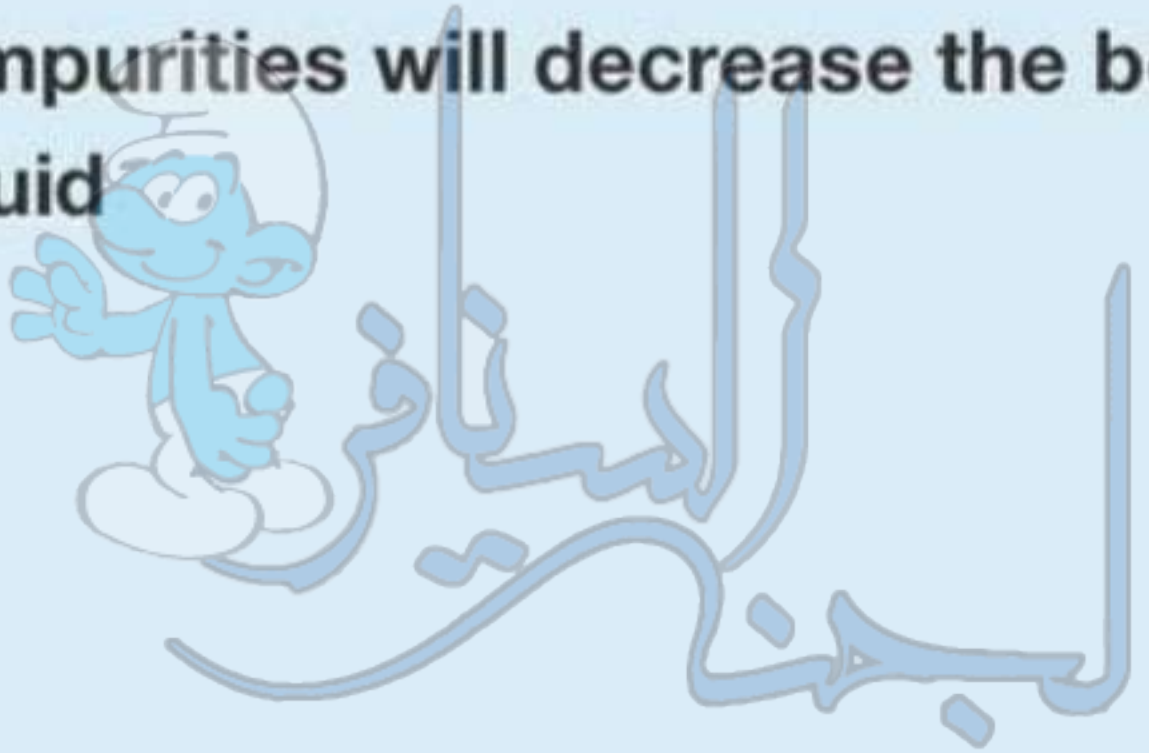
PbI_2 is soluble in water H_2O

اخترأحد الخيارات

- صح
- خطأ



insoluble impurities will decrease the boiling point of liquid



Select one:

- True
- False

If the % mass of H_2O = 27.5%, and the molar mass (Mm) of anhydrous salt = 379 g/mol. The value of water of .crystallization (X) is: (Mm H_2O =18g/mol)

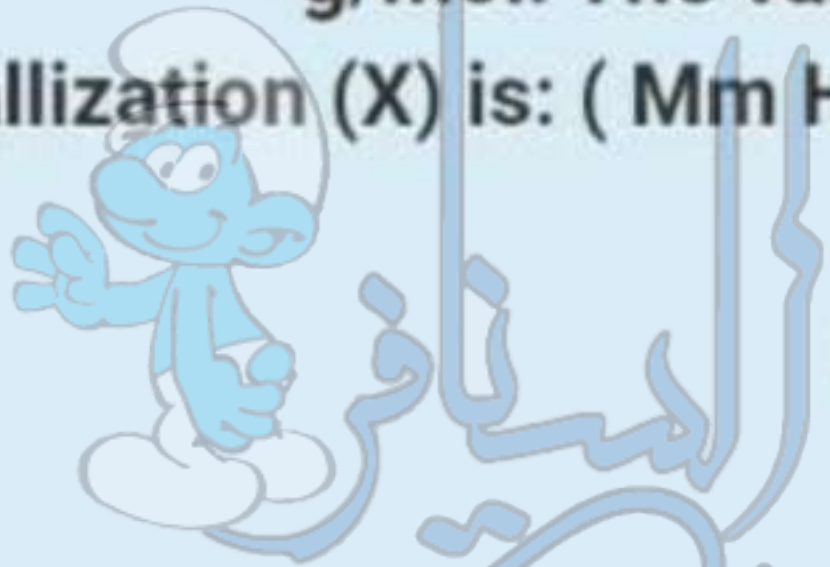
اختر أحد الخيارات

A. 8

B. 10

C. 12

D. 16



Regarding the accuracy of the balances ,which of the following is correct?

Select one:

- A. Top loading balance give mass with accuracy (± 0.001)
- B. Analytical balance give mass with accuracy (± 0.0001)
- C. Top loading balance give mass with accuracy ± 0.0001)
- D. None of the choices

??Which one of these salts is an Alum



اخترأحد الخيارات

A. $KAl(SO_3)_2.12H_2O$

B. $CaSO_4.5H_2O$

C. $NaAl(SO_4)_2.12H_2O$

D. $CaMg(SO_4)_2.10H_2O$

The incorrect statement of the following :is

اخترأحد الخيارات

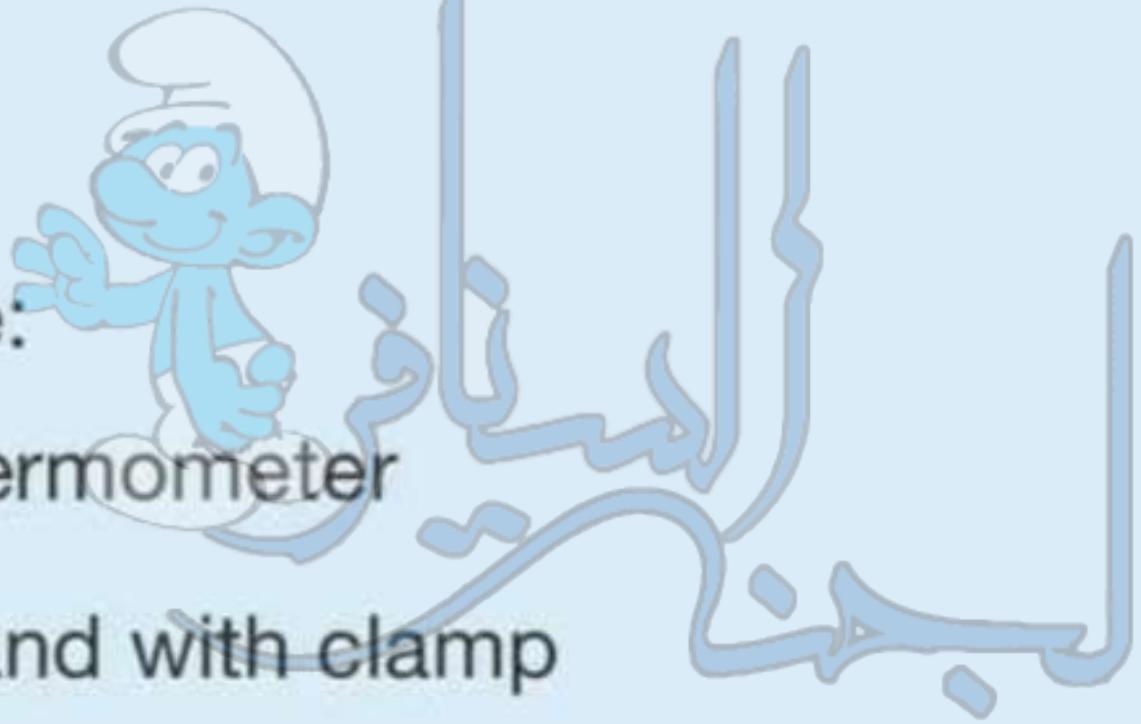
A. To avoid chemical loss, unused chemicals should not be returned to .the stock bottles

B. When dealing with flammable liquid .heat on blue flam directly

C. You can't carry reagent bottles to .your desk and take a sample

D. Always discard waste according to .your instructor order

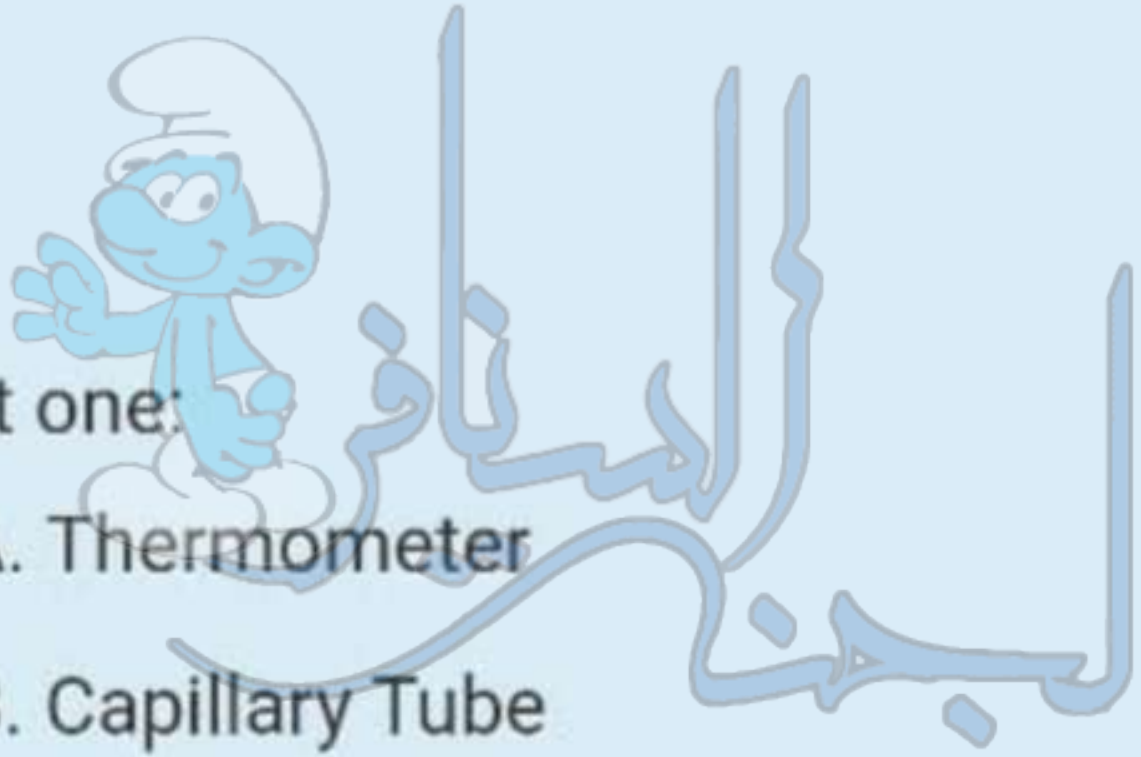
Which of the following apparatus dose not used in boiling point determination set up ?



Select one:

- A. Thermometer
- B. Stand with clamp
- C. Crucible
- D. Capillary Tube

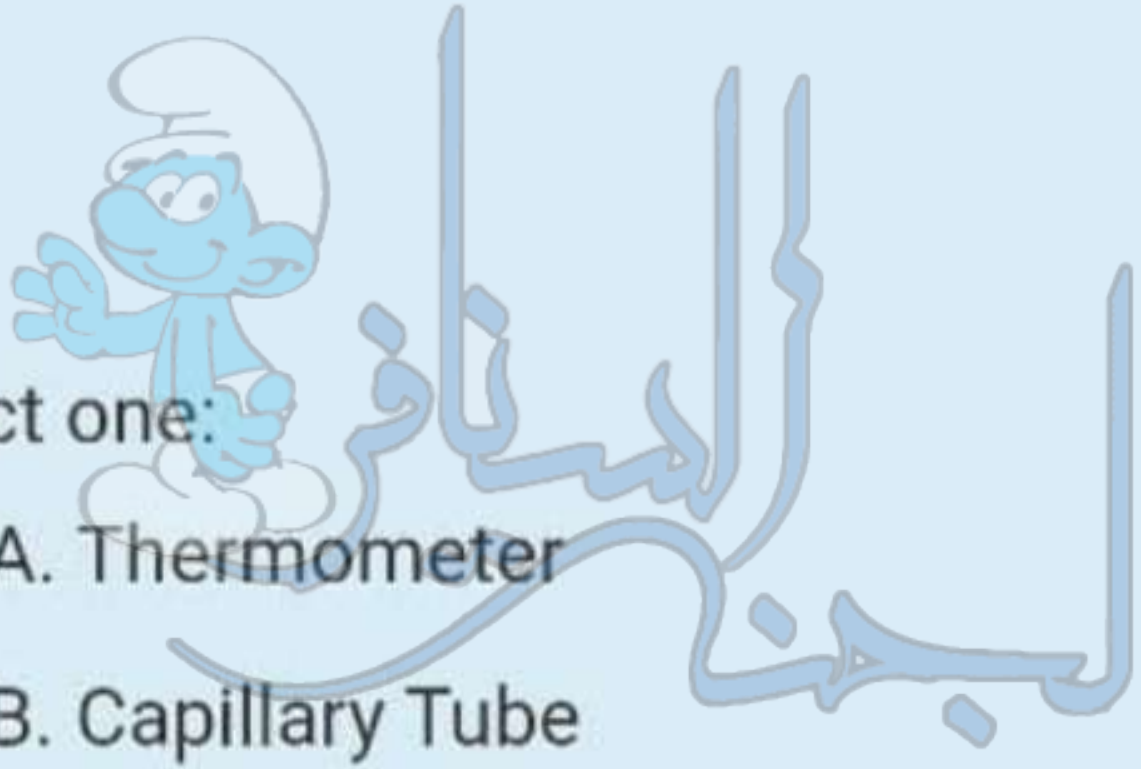
Which of the following apparatus dose not used in boiling point determination set up ?



Select one:

- A. Thermometer
- B. Capillary Tube
- C. Stand with clamp
- D. Flask

Which of the following apparatus dose not used in boiling point determination set up ?



Select one:

- A. Thermometer
- B. Capillary Tube
- C. Stand with clamp
- D. Flask

The incorrect statement of the following is:

اخترأحد الخيارات

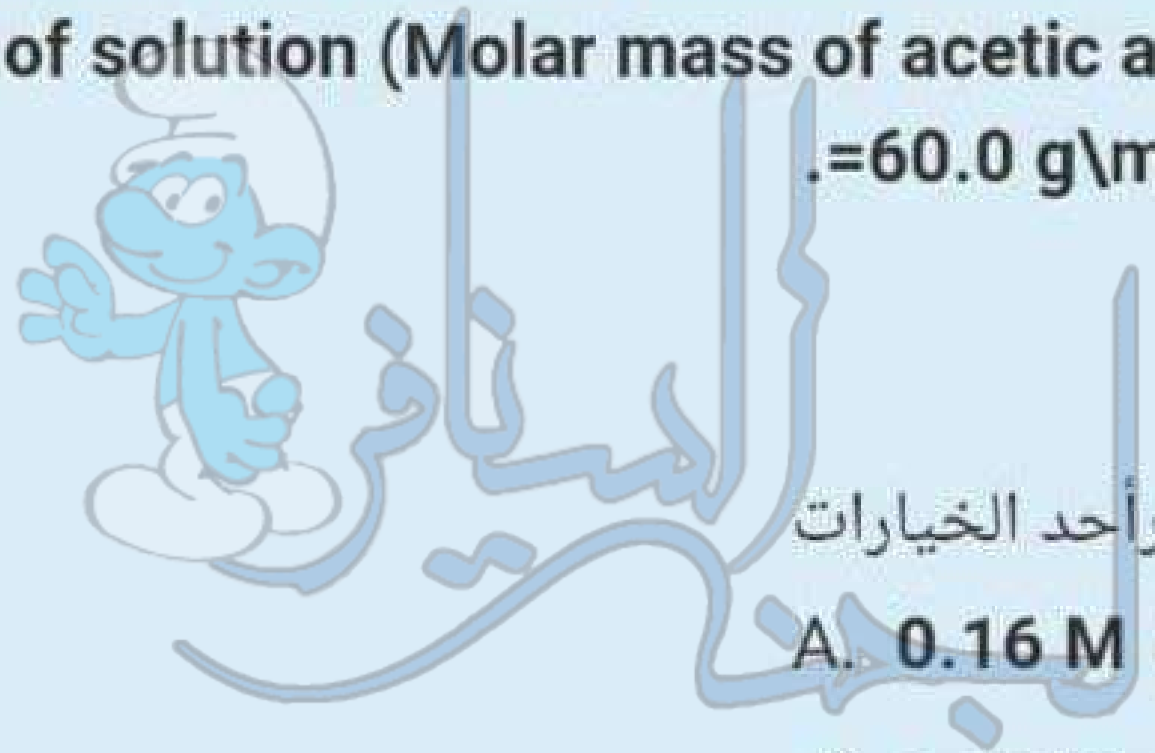
A. To avoid chemical loss, unused chemicals should not be returned to the stock bottles.

B. When dealing with flammable liquid heat on blue flam directly.

C. You can't carry reagent bottles to your desk and take a sample.

D. Always discard waste according to your instructor order.

mL of vinegar solution was titrated 25.0 with 40.0 mL of 0.10M NaOH solution. Calculate the concentration of acetic acid in of solution (Molar mass of acetic acid .=60.0 g\mol)



اختر أحد الخيارات

A. 0.16 M

B. 0.18 M

C. 6.25 M

D. 0.065 M

Sodium hypochlorite NaOCl serve as..... in bleach analysis



Select one:

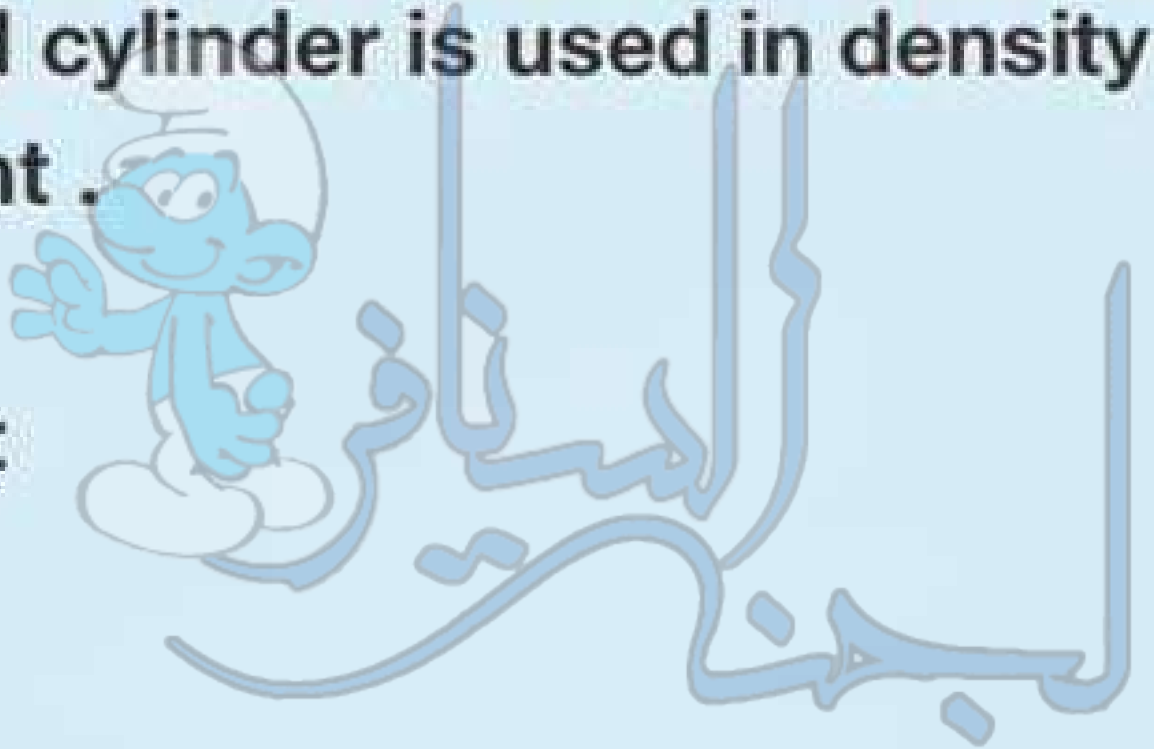
- A. **oxidizing agent**
- B. **reducing agent**

Which of the following is incorrect concerning Limiting reagent?

Select one:

- A. **It's the reactant that determines the amount of product that can be formed in a reaction**
- B. **It's not used up completely in a reaction.**
- C. **It's used up completely in a reaction**
- D. **It's the reactant that used up first.**

Graduated cylinder is used in density of solid Experiment .



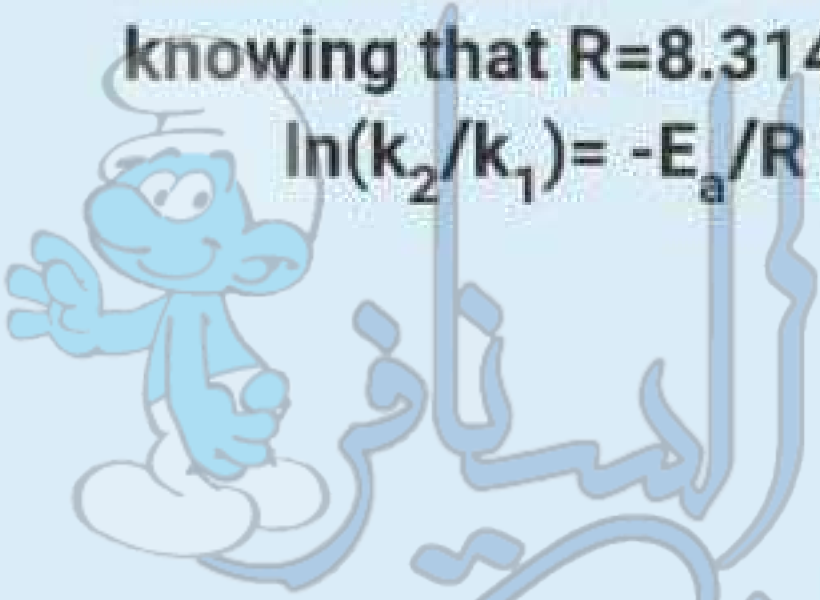
Select one:

- True
- False

If the reaction's rate constant k at 25 °C is 33 $M^{-1}s^{-1}$ and 65 $M^{-1}s^{-1}$ at 402 °C, Calculate the activation energy of the reaction in J/mol .

knowing that R=8.3145 J.K/mol and

$$\ln(k_2/k_1) = -E_a/R\,((1/T_2)-(1/T_1))$$



اخترأحد الخيارات

A. E_a =21.345 J/mol

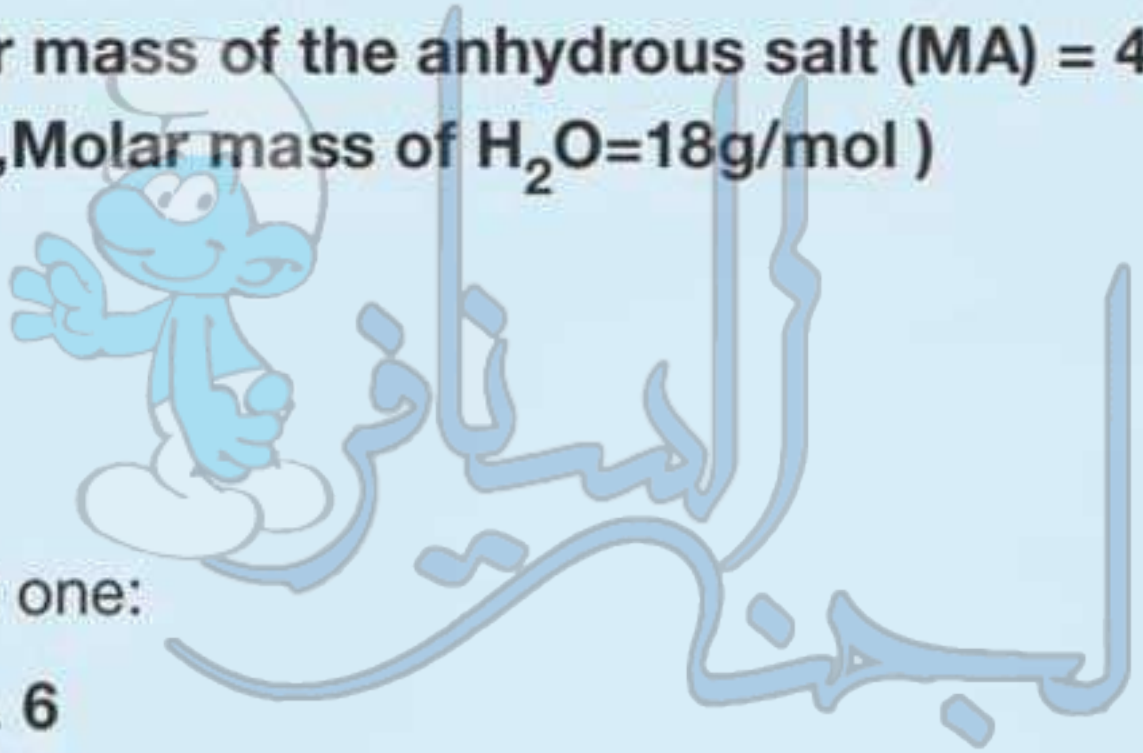
B. E_a = 13.758 J/mol

C. E_a = 2266.1 J/mol

D. E_a =3007.2 J/mol

The water of crystallization (X) for an unknown hydrate that is found to contain 18 % water is equal :

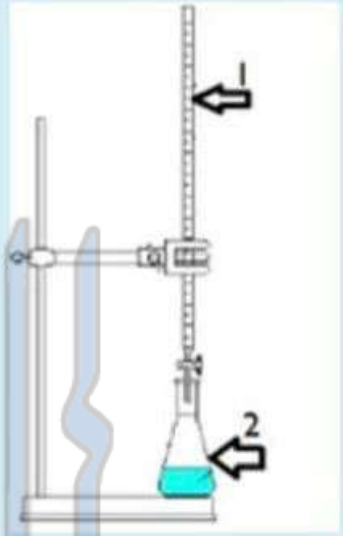
(molar mass of the anhydrous salt (MA) = 490 g /mol ,Molar mass of H_2O=18g/mol)



Select one:

- A. **6**
- B. **8**
- C. **3**
- D. **15**

Consider the set up on the right which used during vinegar analysis. The names of solution filled in equipment 1 and equipment 2 are:



Select one:

- a. in 1 NaOH solution and in 2 CH_3COOH (vinegar solution)
- b. in 1 CH_3COOH (vinegar solution) and in 2 HCl solution
- c. in 1 CH_3COOH (vinegar solution) and in 2 NaOH solution
- d. in 1 HCl solution and in 2 NaOH solution

For the reaction $A+B \rightarrow C+D$. If the order of reactant A and B is 1 and 0 respectively ;which unit represent the rate law constant K unit ?

Select one:

- A. $L\ mol^{-1}\ S^{-1}$
- B. S^{-1}
- C. $mol^{-1}\ L^{2}\ S^{-1}$
- D. $mol\ L^{-1}\ S^{-1}$

A 10.0mL bleach sample is diluted to 100mL in a volumetric flask. A 25mL of this solution is analyzed according to the procedure you learned .If 11.3mL of 0.30M $Na_2S_2O_3$ is needed to reach the end point, calculate the mass percent of NaClO(M.m=74.50g/mol) in the original sample? (Assume the density of bleach solution is 1.084g\mL).

Select one:

- A. **2.4 %**
- B. **0.79 %**
- C. **5.2 %**
- D. **4.7 %**

The vapor of an unknown volatile liquid occupies a 269mL Erlenmeyer flask at 98.7°C and 748 torr. The mass of vapor is 0.791g. What is the molar mass of ? thisvolatile liquid in g/mol

(1atm =760torr /R=0.08206 L.atm/K.mol)



اخترأحد الخيارات

A. 122.52

B. 93

C. 91.1

D. 88.23

Nitric acid (HNO_3) can be neutralized by any base to form a salt and water, as in the following equation:

$$Mg(OH)_2 + 2\,HNO_3 \rightarrow Mg(NO_3)_2 + 2\,H_2O$$

What is the mass of $Mg(NO_3)_2$ (M.m=148.33g/mol) will be formed when 250.0 g $Mg(OH)_2$(M.m=58.35g/mol) reacted with 1000g HNO_3(M.m= 63.02g/mol)?

Select one:

- A. **294 g**
- B. **318 g**
- C. **147 g**
- D. **636 g**

The water of crystallization (X) for an unknown hydrate that is found to contain 18 % water is equal :

molar mass of the anhydrous salt (MA) =) 490 g /mol ,Molar mass of H_2O=18g/mol (

اخترأحد الخيارات

A. 3

B. 15

C. 8

D. 6

use the following table to calculate the over all order of the reaction $2I^- + H_2O_2 + H^+ \rightarrow I_2 + 2H_2O$

Trial	$[I^-]$	$[H_2O_2]$	Initial rate
1	0.2	0.2	0.4
2	0.2	0.4	0.8
3	0.4	0.2	0.8

Select one:

- A. **Third order**
- B. **Zero order**
- C. **first order**
- D. **Second order**

When should the starch indicator be added in the bleach titration ?

Select one:

- A. **At the beginning of titration**
- B. **When the color of solution become pale yellow**
- C. **When the color of solution become colorless**
- D. **When I_3^- is formed**

The name and color of precipitate that result during limiting reactant experiment is:



Select one:

- A. $CaCl_2$, white
- B. $CaCO_3$, pink
- C. $CaCO_3$, White
- D. $CaCl_2$, pink

calculate weight of acetic acid at end point if the concentration of NaOH =0.01 M and needs 5 ml to reach end point .molar mass of acetic acid =60 g\mol

Select one:

- a. 0.06
- b. 3*10^-3
- c. 6*10^-3
- d. 0.03

Which of the following is not a property of a volatile compound?

Select one:

- A. They have high boiling points
- B. They have weak intermolecular interactions
- C. They are heated in water bath not over direct flame because they are usually flammable
- D. They have high vapor pressure
- E. Their molecular weight is determined by Dumas's method

What is the formula to calculate heat energy required to raise the temperature of any substance?

اخترأحد الخيارات

A.
Q=m c

B.
Q= m s Δt

C.
Q= ½ m v

D.
m=Q C

energy in exothermic reactions are

Select one:

- a. released
- b. absorbed
- c. ended

Given the following set of data:

Volume of 2.10M HCl (in the cup)	**50.00mL**
Volume of 2.10M NaOH added	**50.00mL**
Initial temperature of HCl and NaOH solutions	**22.0°C**
Final temperature after mixing solutions	**28.6°C**
Heat capacity of the cup	**95.0J\°C**
density of HCl and NaOH solutions	**1.00g\mL**
Specific heat of the produced solution	**4.10 J\g.°C**

Calculate ΔH for the neutralization reaction of HCl and NaOH, in kJ\mol H_2O produced.

Select one:

- A. **-50.6**
- B. **-54.4**
- C. **-31.7**
- D. **-28.1**

All nitrates(NO_3) are water soluble and all ammonium(NH_4) salts too.

Select one:

- True
- False

A Chrome alum ($KCr(SO_4)_2.12H_2O$),(Mm alum =499g/mol), Calculate the mass of water in .12.9 g sample of this alum: (Mm H_2O =18g/mol)

اختر احد الخيارات

A. 7.02 g

B. 2.92 g

C. 3.98 g

D. 5.58 g

um sulfate hydrate salt ,($CaSO_4.XH_2O$), is heated in a crucible until a constant mass is reached. If the mass of of .anhydrous $CaSO_4$ salt equal 1.426g.Calculate the percent by mass of water in the hydrated calcium sulfate

(M.mass : H_2O=18g/mol ,$CaSO_4$ =136.1g/mol)

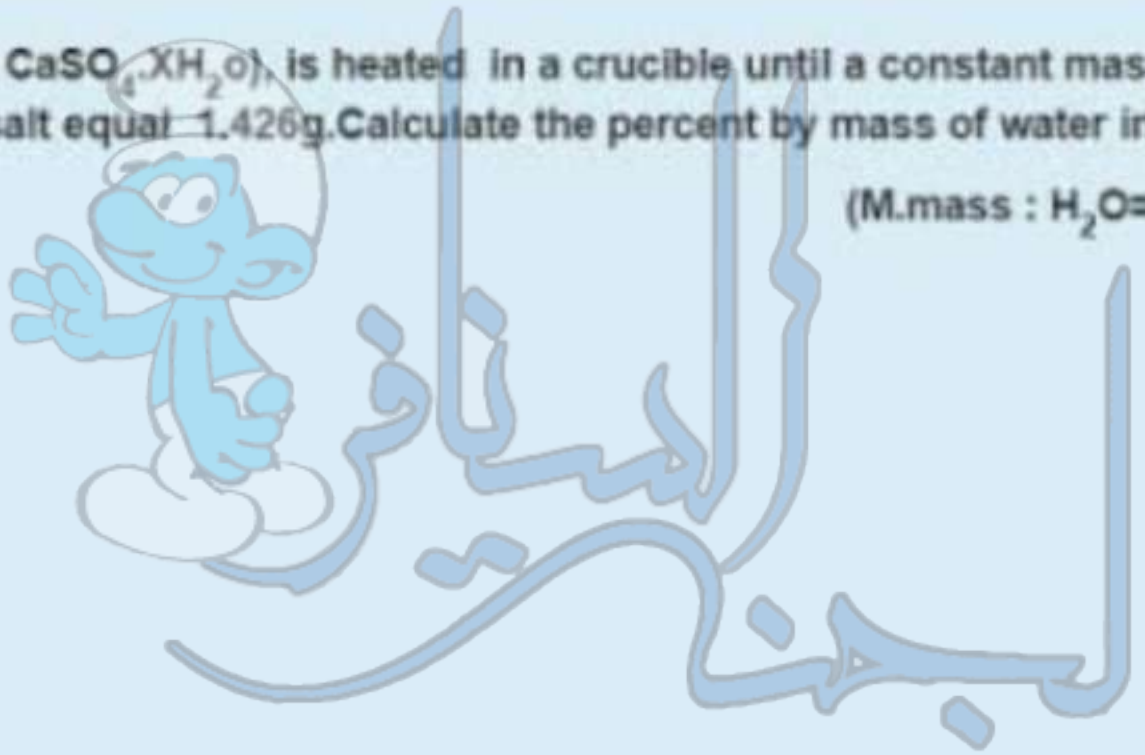
اختر احد الخيارات

% A. 20.9

% B. 0.79

% C. 2.09

% D. 79.0



Which of the following apparatus dose not used in boiling point determination set up

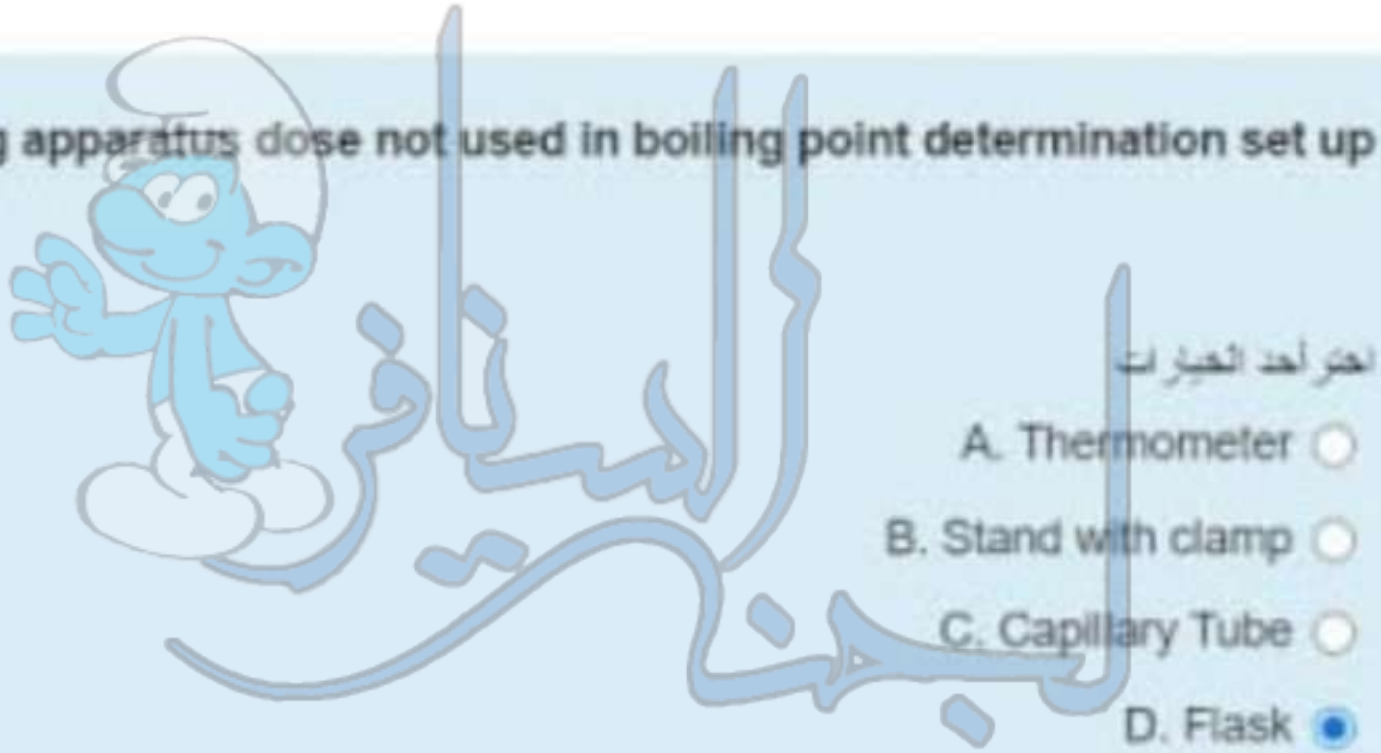
اختر أحد الخيارات

A. Thermometer

B. Stand with clamp

C. Capillary Tube

D. Flask



?? What is the name of the following glassware



اختر احد الخيارات

a. Graduated cylinder ○

b. Erlenmeyer flask ○

c. Volumetric flask ○

d. Beaker ●

??Which one of these salts is an Alum

اختر احد الخيارات

A. $NaAl(SO_4)_2.12H_2O$

B. $CaMg(SO_4)_2.10H_2O$

C. $KAl(SO_3)_2.12H_2O$

D. $CaSO_4.5H_2O$



?Regarding the accuracy of the balances ,which of the following is correct

اختر احد الخيارات

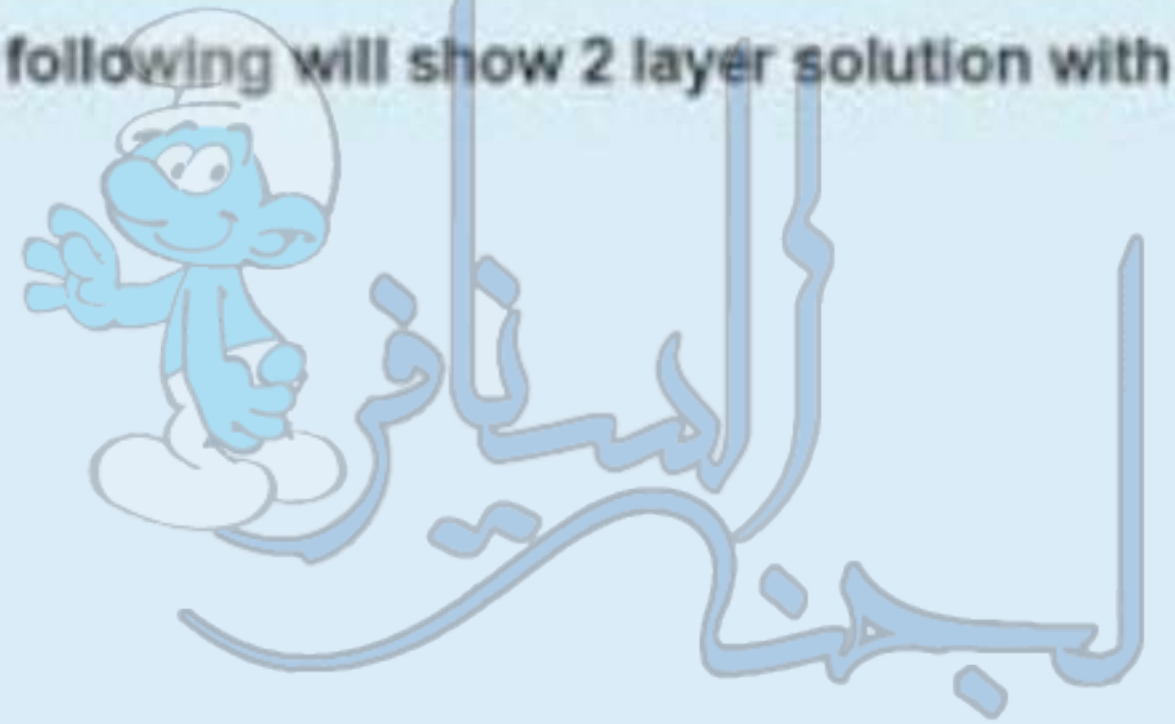
A. Top loading balance give mass with accuracy (± 0.001)

B. Analytical balance give mass with accuracy (± 0.0001)

(C. Top loading balance give mass with accuracy ± 0.0001

D. None of the choices

?Which of the following will show 2 layer solution with Hexane (C_6H_{14})



اختر أحد الخيارات

A. C_6H_6

B. C_4H_{10}

C. OII

D. HCl

:The solubility is defined as

اختر احد الخيارات

A. The amount of material that dissolves in a given amount of solid

B. The amount of solvent that dissolves in a given amount of solute

C. The amount of material that dissolves in a given amount of solution

D. The amount of material that dissolves in a given amount of solvent

?Which of the following statement is correct

اختر احد الخيارات

a. $CH_3CH_2CH_2OH$ have higher boiling point than $CH_3CH_2CH_2CH_2CH_2OH$

b. CH_3OH have higher boiling point than CH_3CH_2OH

c. $CH_3CH_2CH_2OH$ have higher boiling point than $CH_3CH_2CH_3$

d. $CH_3CH_2CH_2OH$ have lower boiling point than $CH_3CH_2CH_3$

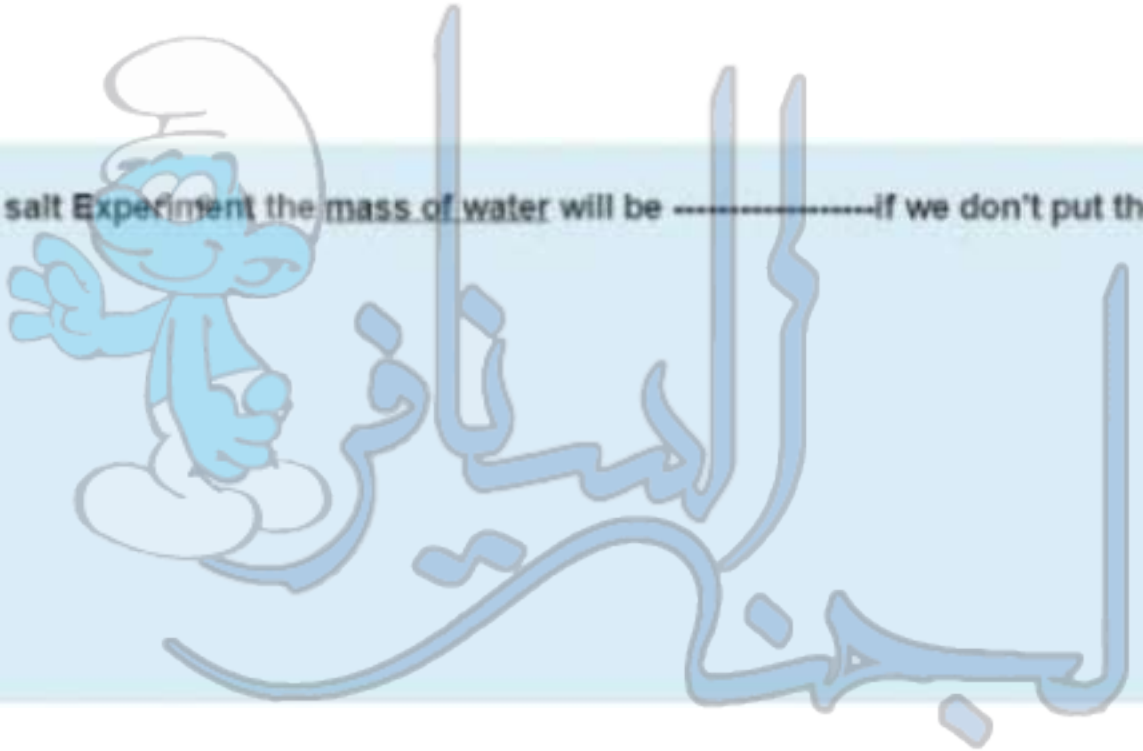
?During the hydrated salt Experiment the mass of water will be ----------if we don't put the sample in desiccator to cool

اختر أحد الخيارات

A. Increase ○

B. Decrease ●

C. not effected ○



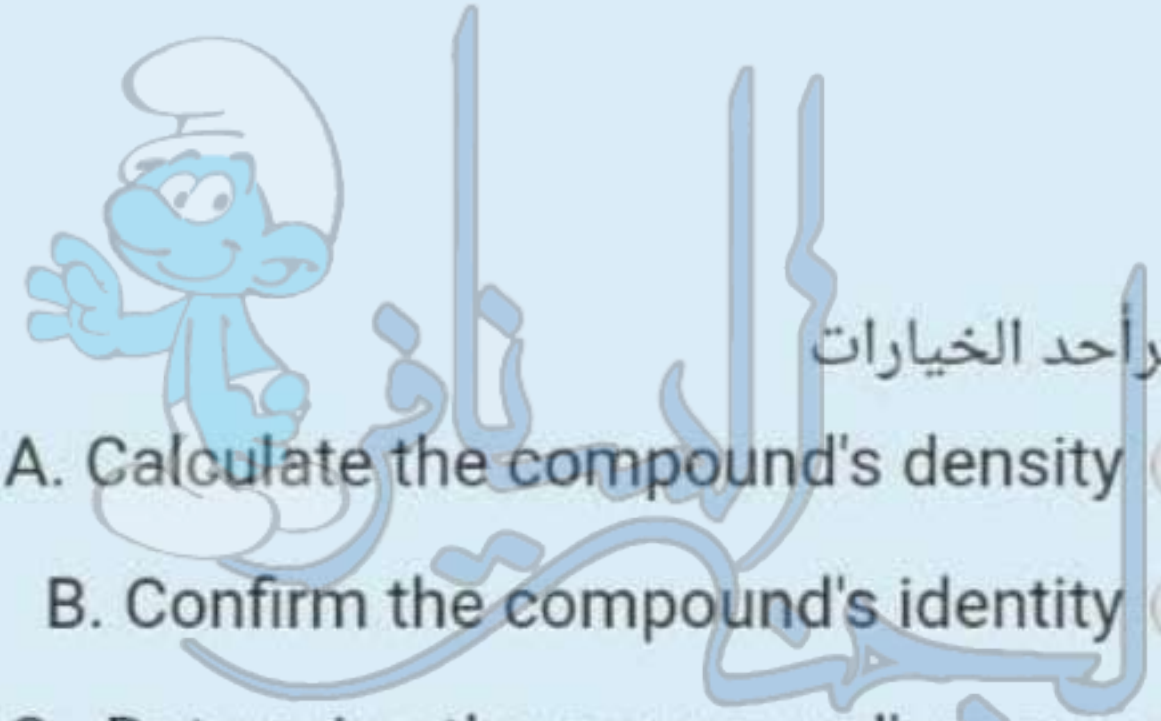
PbI_2 is soluble in water H_2O



اختراحد الخيارات

- صح
- خطأ

Measuring the boiling point of a compound can be used to :



اختر أحد الخيارات

A. Calculate the compound's density

B. Confirm the compound's identity

C. Determine the compound's vapor pressure

D. Determine the compound's structure

A compound is a hydrate when its absorbed water from out side

Select one:

- True
- False



insoluble impurities Dose not effect the boiling point of liquid



اخترأحد الخيارات

- صح
- خطأ

PbI_2 is soluble in water H_2O



اخترأحد الخيارات

- صح
- خطأ

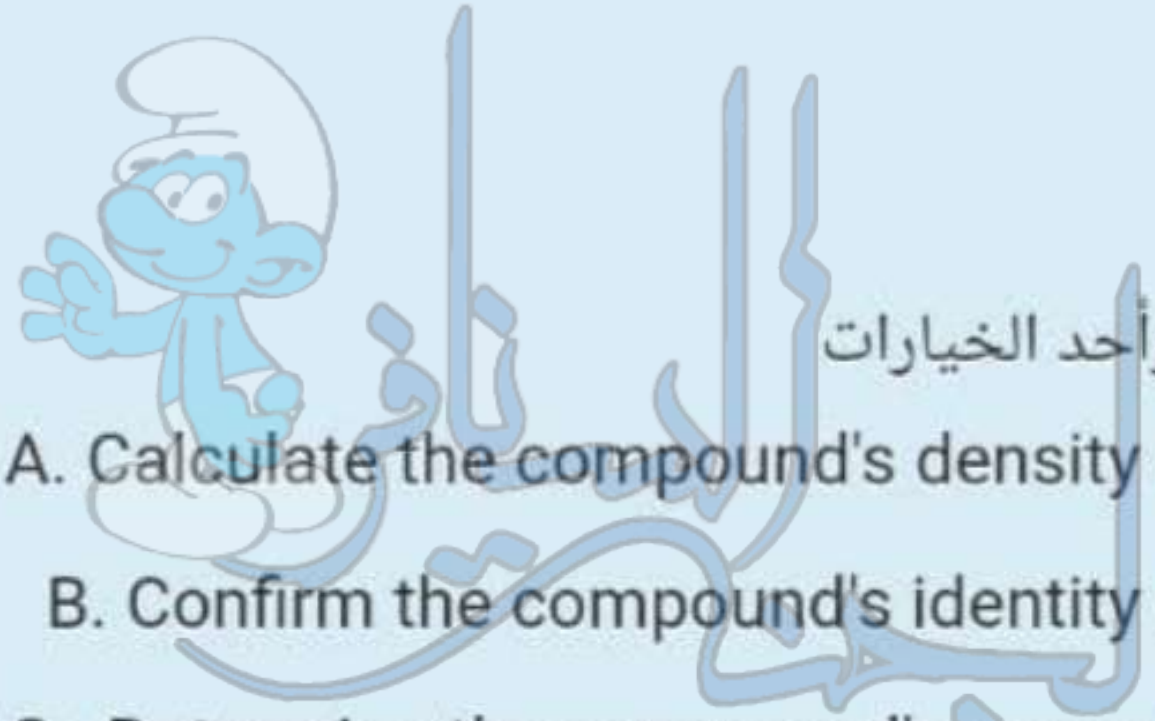
Which of the following will show 2 layer solution with Hexane (C_6H_{14})?

Select one:

- A. Oil
- B. HCl
- C. C_6H_6
- D. C_4H_{10}



Measuring the boiling point of a compound can be used to :



اختر أحد الخيارات

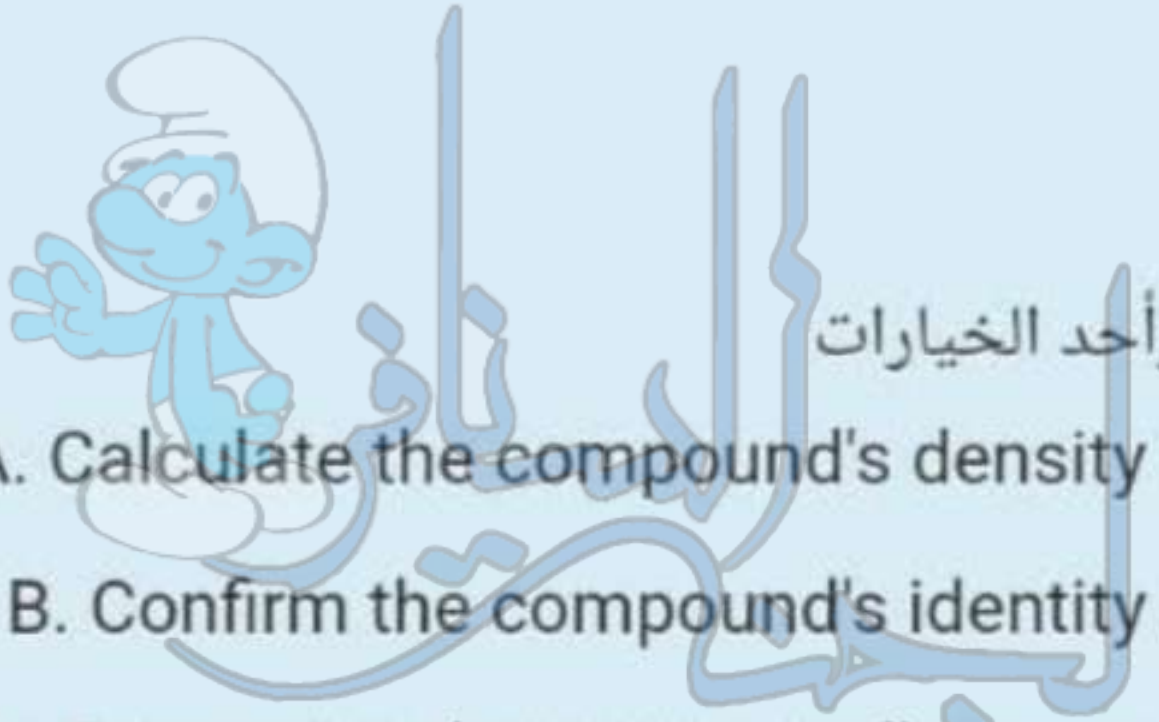
A. Calculate the compound's density

B. Confirm the compound's identity

C. Determine the compound's vapor pressure

D. Determine the compound's structure

Measuring the boiling point of a compound can be used to :



اختر أحد الخيارات

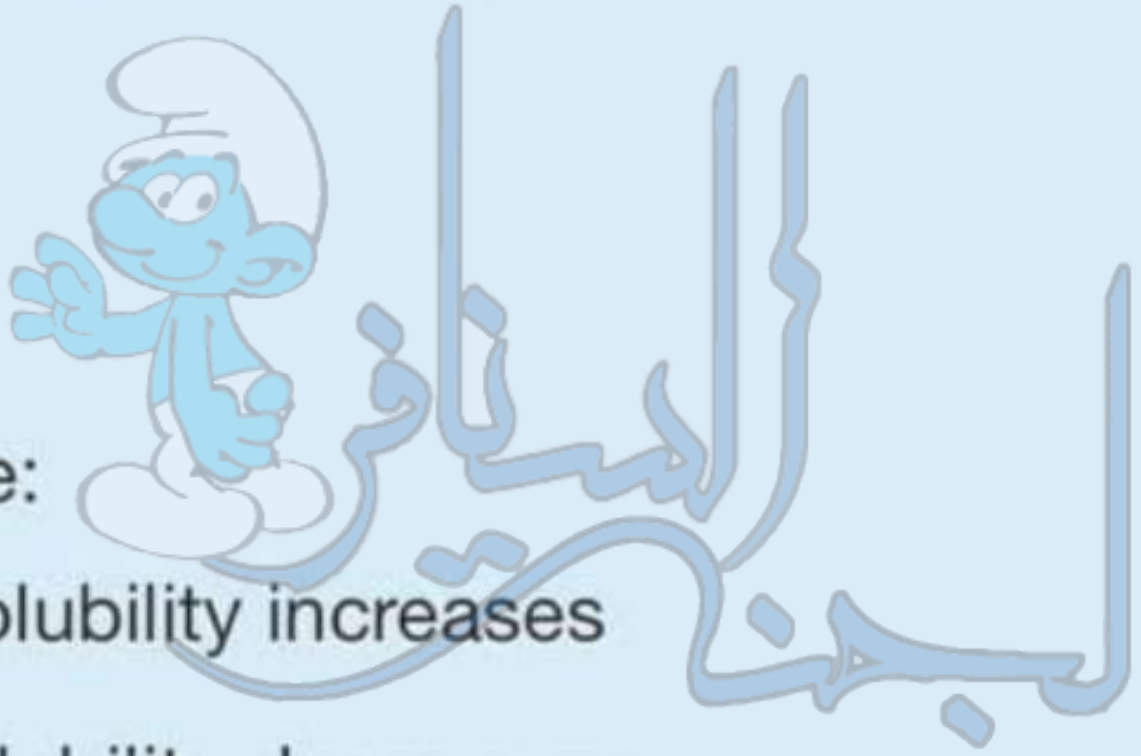
A. Calculate the compound's density

B. Confirm the compound's identity

C. Determine the compound's vapor pressure

D. Determine the compound's structure

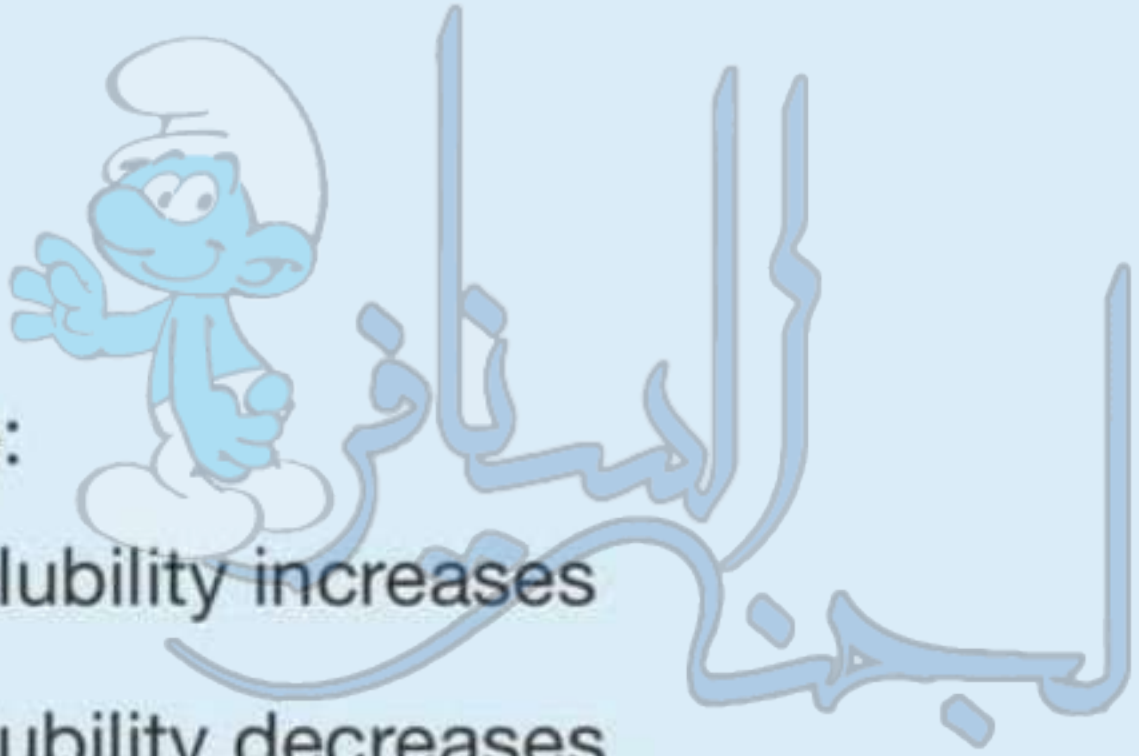
For most of the solid solutes, What happens to the solubility if the temperature decreases?



Select one:

- A. Solubility increases
- B. Solubility decreases
- C. Solubility stays the same
- D. None of the choices

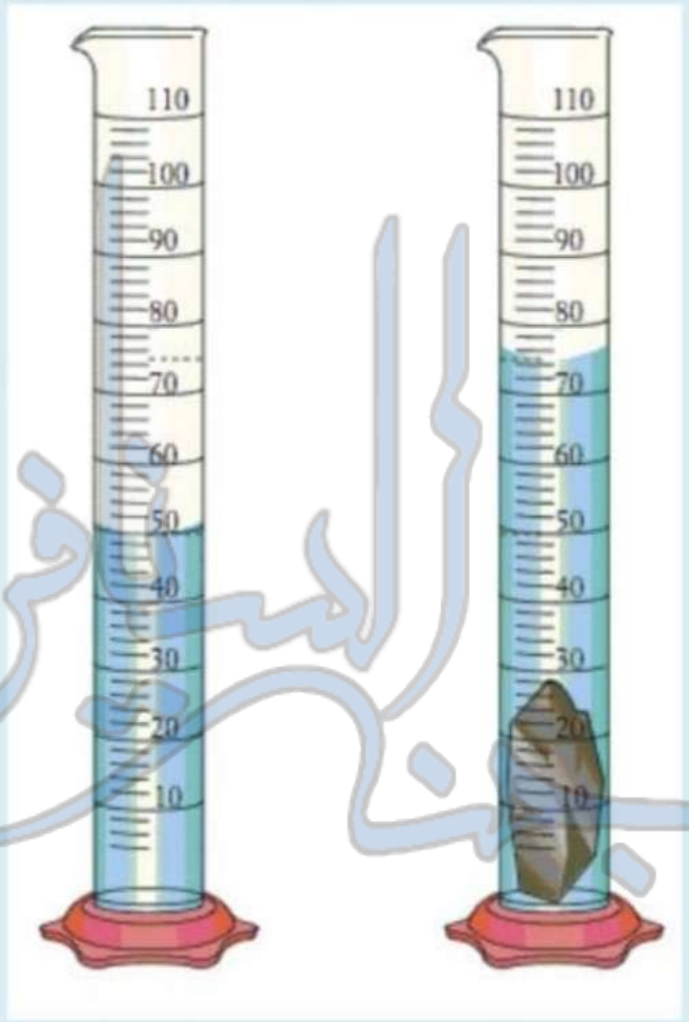
For most of the solid solutes, What happens to the solubility if the temperature decreases?



Select one:

- A. Solubility increases
- B. Solubility decreases
- C. Solubility stays the same
- D. None of the choices

The Density of material with 0.13 Kg weight equal _____, if during experiment the water level was according the photo beside:



Select one:

A. 5.2 X 10^{-3} g/mL

B. 52 X 10^{-3} g/mL

C. 0.052 g/mL

D. 5.2 g/mL

What is the name of the following glassware ??



Select one:

- a. Volumetric flask
- b. Burette
- c. Erlenmeyer flask
- d. Beaker

During the hydrated salt Experiment the mass of water will be ----------if we use red flame for heating the crucible and its content.



اخترأحد الخيارات

A. Decrease

B. Increase

C. not effected

The correct statement of the following is:

Select one:

- A. All chemicals are hazardous.
- B. Eating, drinking and phoning are allowed in the lab.
- C. To avoid chemical loss, unused chemicals should be returned to the stock bottles.
- D. You can do your lab experiment without wearing lab coat.

Which of the following apparatus dose not used in boiling point determination set up ?



اخترأحد الخيارات

A. Capillary Tube

B. Stand with clamp

C. Crucible

D. Thermometer

Which equipment is used to measure the volume of a solid in density determination experiment?



اخترأحد الخيارات

A. Graduated cylinder

B. Thermometer

C. Balance

D. flask

All the following is true regarding the boiling point of liquid except?

Select one:

- A. As the external pressure increases the boiling point increases
- B. As the temperature increases the vapor pressure of liquid decreases
- C. It is used to check purity of compound
- D. Temperature at which vapor pressure of liquid exactly equals the external pressure

A Chrome alum ($KCr(SO_4)_2.12H_2O$),(Mm alum =499g/mol), Calculate the mass of water in 12.9 g sample of this alum: (Mm H_2O =18g/mol).

اخترأحد الخيارات

A. 2.92 g

B. 7.02 g

C. 5.58 g

D. 3.98 g

Which of the following pressure will give you the lowest boiling point of ethanol:

Select one:

- A. 1 atm
- B. 0.92 atm
- C. 2.5 atm
- D. 1.95 atm



The type of reaction in acid base experiment is

Select one:

- A. **Precipitation reaction**
- B. **Direct titration reaction**
- C. **Redox reaction**
- D. **Back titration reaction**

Nitric acid (HNO_3) can be neutralized by any base to form a salt and water, as in the following equation:

$$Mg(OH)_2 + 2\,HNO_3 \rightarrow Mg(NO_3)_2 + 2\,H_2O$$

What is the mass of $Mg(NO_3)_2$ (M.m=148.33g/mol) will be formed when 250.0 g $Mg(OH)_2$(M.m=58.35g/mol) reacted with 1000g HNO_3(M.m= 63.02g/mol)?

Select one:

- A. 294 g
- B. 318 g
- C. 147 g
- D. 636 g

Endothermic reactions feel

اختر أحد الخيارات

a. warm

b. cold

The W/W % of NaClO in a stock bleach solution is 7%. The molarity of diluted solution of this solution is 0.037 M. Given that the molar mass of NaClO= 74.4g/mol and the density of the stock solution = 1.1 g/mL. Calculate the dilution factor for diluted sample?

اختر أحد الخيارات

A. 28

B. 18

C. 14

D. 5

Which of the following is correct for an exothermic reaction ?

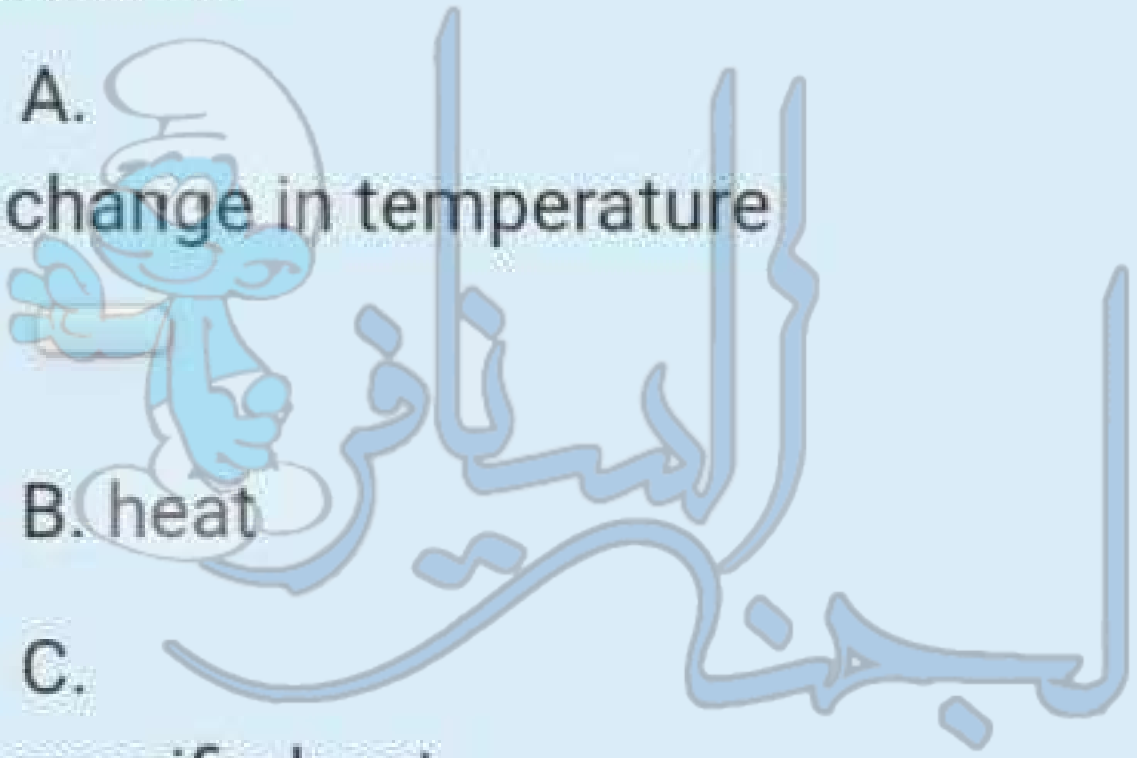
Select one:

- A. ΔH_{system} is +ve
- B. ΔH_{system} is -ve
- C. $\Delta H_{surrounding\ and\ system}$ is +ve
- D. $\Delta H_{surrounding}$ is +ve

q stands for which variable

Select one:

- A. change in temperature
- B. heat
- C. specific heat
- D. mass



Calculate the enthalpy of the following chemical reaction:

$$CS_2(\ell) + 3O_2(g) \longrightarrow CO_2(g) + 2SO_2(g)$$

Given:

$$C(s) + O_2(g) \longrightarrow CO_2(g)\ \Delta H = -393.5\ kJ/mol$$
$$S(s) + O_2(g) \longrightarrow SO_2(g)\ \Delta H = -296.8\ kJ/mol$$
$$C(s) + 2S(s) \longrightarrow CS_2(\ell)\ \Delta H = -87.9\ kJ/mol$$

Select one:

- a. -898.7 KJ
- b. +112.7KJ
- c. +1074.5 KJ
- d. -1074.5 KJ

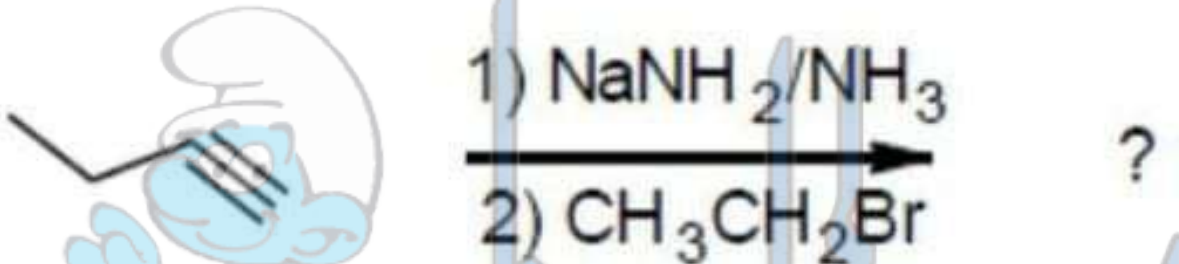
What is the molecular formula for a compound with the empirical formula of$CaCl_2$ (M.m=110.9g/mol) and a molecular mass of 330g/mol.



Select one:

- A. Ca_2Cl_4
- B. Ca_3Cl_9
- C. Ca_3Cl_6
- D. Ca_3Cl_5

What is the IUPAC name of the product ?that resulted from the reaction



اخترأحد الخيارات

- hexyne-1
- hexyne-2
- diethylethyne-1,2
- ethylbutyne-1
- hexyne-3

Use the data on the table to calculate the % NaOCl in original bleach sample

Mass of original bleach	2.3 g
Volume of bleach solution	25 mL
Initial burette reading of $Na_2S_2O_3$	2.5 mL
Final burette reading of $Na_2S_2O_3$	19.0 mL
Molarity of $Na_2S_2O_3$	0.15 M
Mm NaOCl	74.5 g/mol

Select one:

- A. 3.7 %
- B. 2.9 %
- C. 5.4 %
- D. 4.0 %

What unit do you use to measure Thermal Energy?

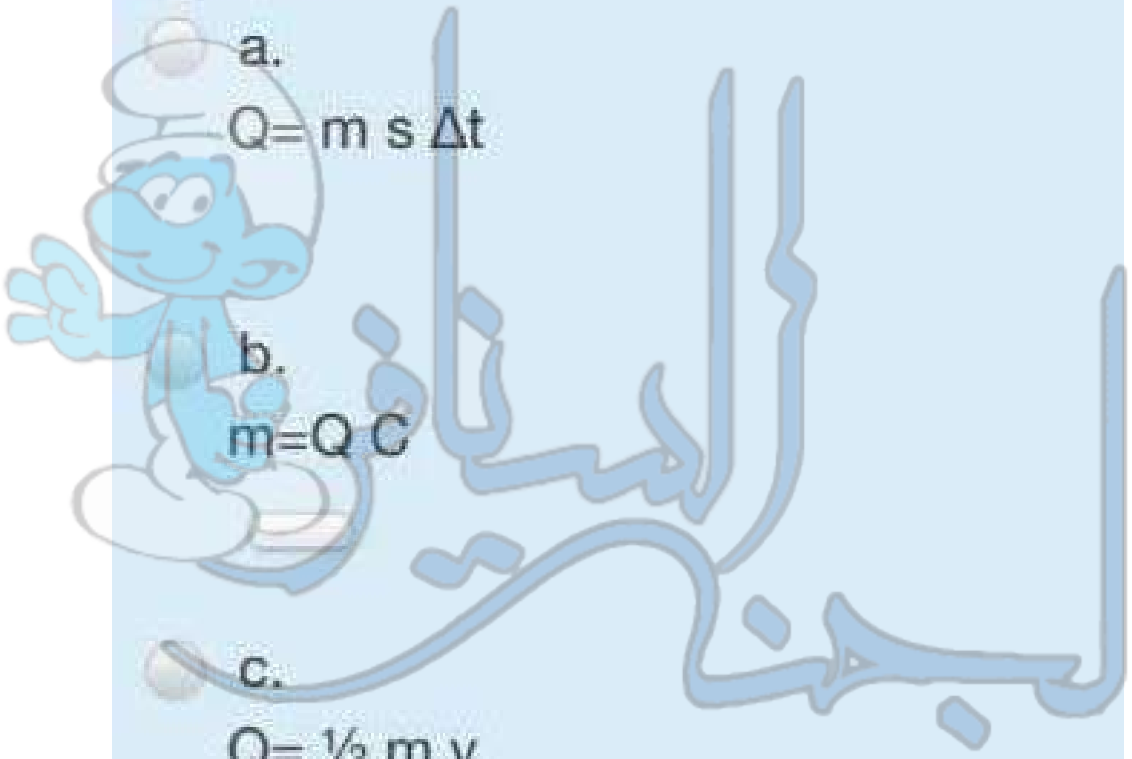
Select one:

a.
$Q = m\ s\ \Delta t$

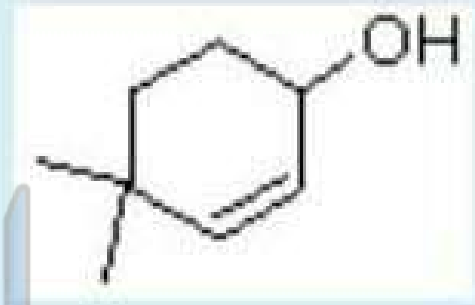
b.
$m = Q\ C$

c.
$Q = \frac{1}{2}\ m\ v$

d.
$Q = m\ c$



What is the correct name for the following compound?



اختر أحد الخيارات

- dimethyl-2-cyclohexen-1-ol-4,4
- dimethyl-1-cyclohexen-3-ol-6,6
- dimethyl-5-cyclohexen-1-ol-4,4
- dimethyl-1-cyclohexen-6-ol-3,3
- dimethyl-2-cyclohexen-4-ol-1,1

How many Significant Figure : in the number

0.0030020

اختر أحد الخيارات

a. 4

b. 5

c. 8

d. 2



:This pictograms indicates that the chemical compound is

اختر أحد الخيارات

A. Explosive

B. Corrosive

C. Toxic

D. flammable

?All the following is true regarding the boiling point of liquid except

اختر أحد الخيارات

A. Temperature at which vapor pressure of liquid exactly equals the external pressure

B. It is used to check purity of compound

C. As the external pressure increases the boiling point increases

D. As the temperature increases the vapor pressure of liquid decreases

?Which one of the following will dissolve in Hexane (C_6H_{14})



اختر أحد الخيارات

A. NaCl

B. I_2

C. H_2O

D. CH_3OH

If the % mass of H_2O = 27.5%, and the molar mass (Mm) of anhydrous salt = 379 g/mol. The value of water of crystallization (X) is: (Mm H_2O =18g/mol).

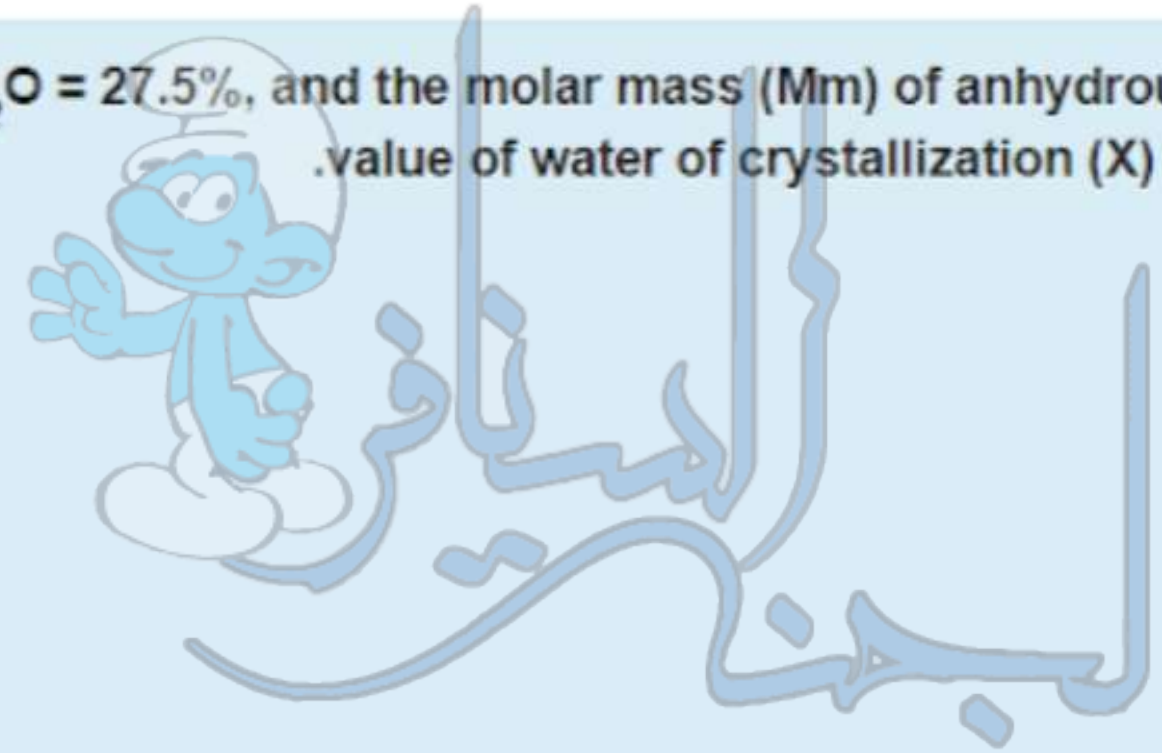
اختر أحد الخيارات

A. 10

B. 12

C. 16

D. 8



?For most of the gases solutes, What happens to the solubility if the pressure decreases



اختر أحد الخيارات

A. Solubility increases

B. Solubility decreases

C. Solubility stays the same

D. None of the choices



اختر أحد الخيارات

a. Beaker

b. Volumetric flask

c. Graduated cylinder

d. Erlenmeyer flask

?Regarding the accuracy of the balances ,which of the following is correct



اختر أحد الخيارات

A. Top loading balance give mass with accuracy (± 0.01) ◉

B. Analytical balance give mass with accuracy (± 0.001) ○

C. Top loading balance give mass with accuracy (± 0.001) ○

D. None of the choices ○

Regarding the accuracy of the balances ,which of the following is correct



اختر أحد الخيارات

A. Top loading balance give mass with accuracy (± 0.01)

B. Analytical balance give mass with accuracy (± 0.0

C. Top loading balance give mass with accuracy (± 0.0

D. None of the choices

?Which equipment is used to measure the mass of a liquid in density experiment

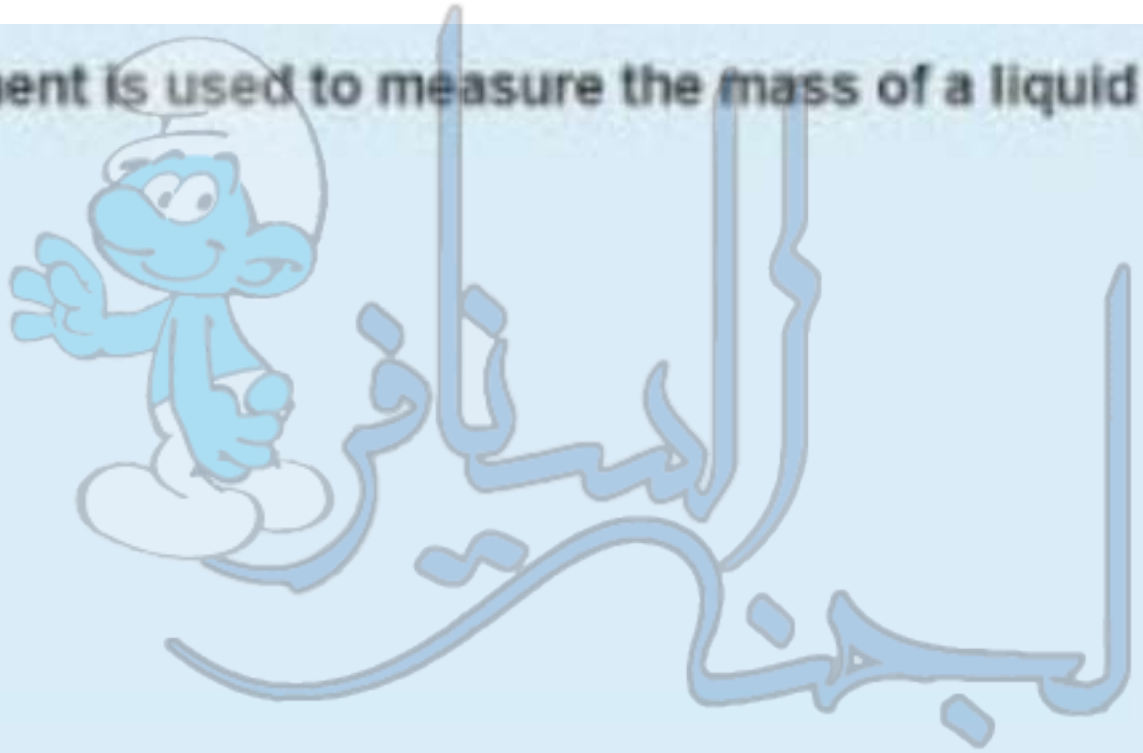
اختر أحد الحيارات

A. Test tube

B. Thermometer

C. Balance

D. Capillary tube



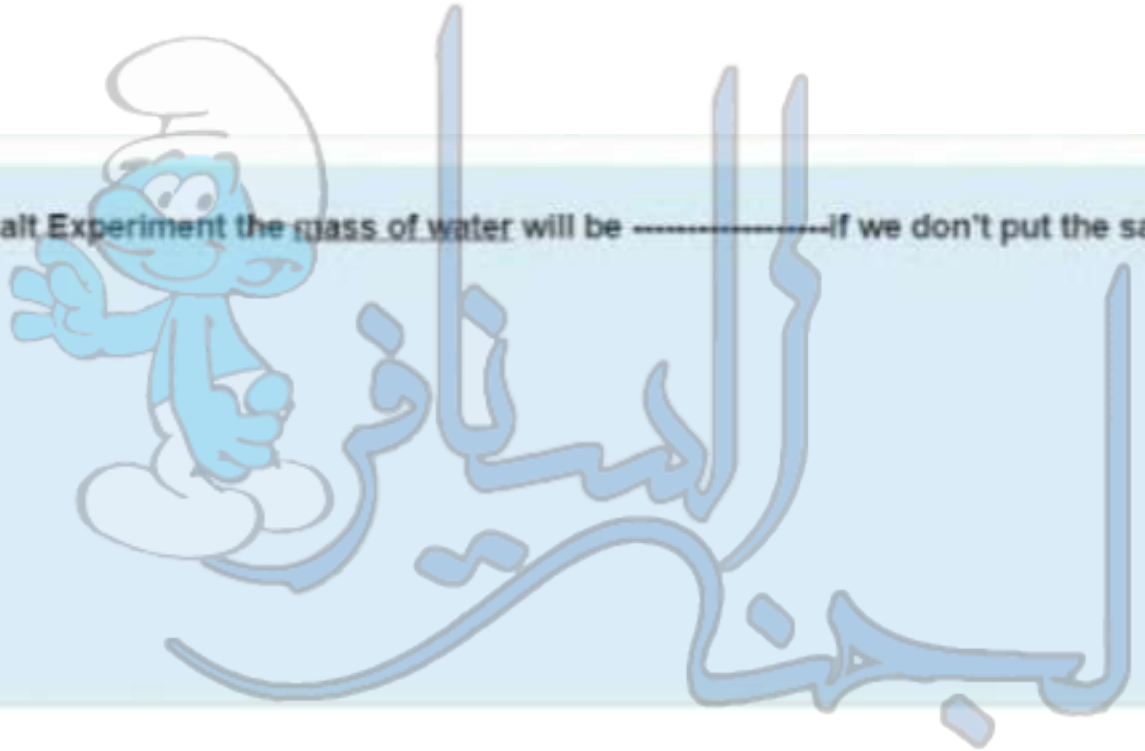
?During the hydrated salt Experiment the mass of water will be ----------if we don't put the sample in desiccator to cool

اختر أحد الخيارات

A. Increase

B. Decrease

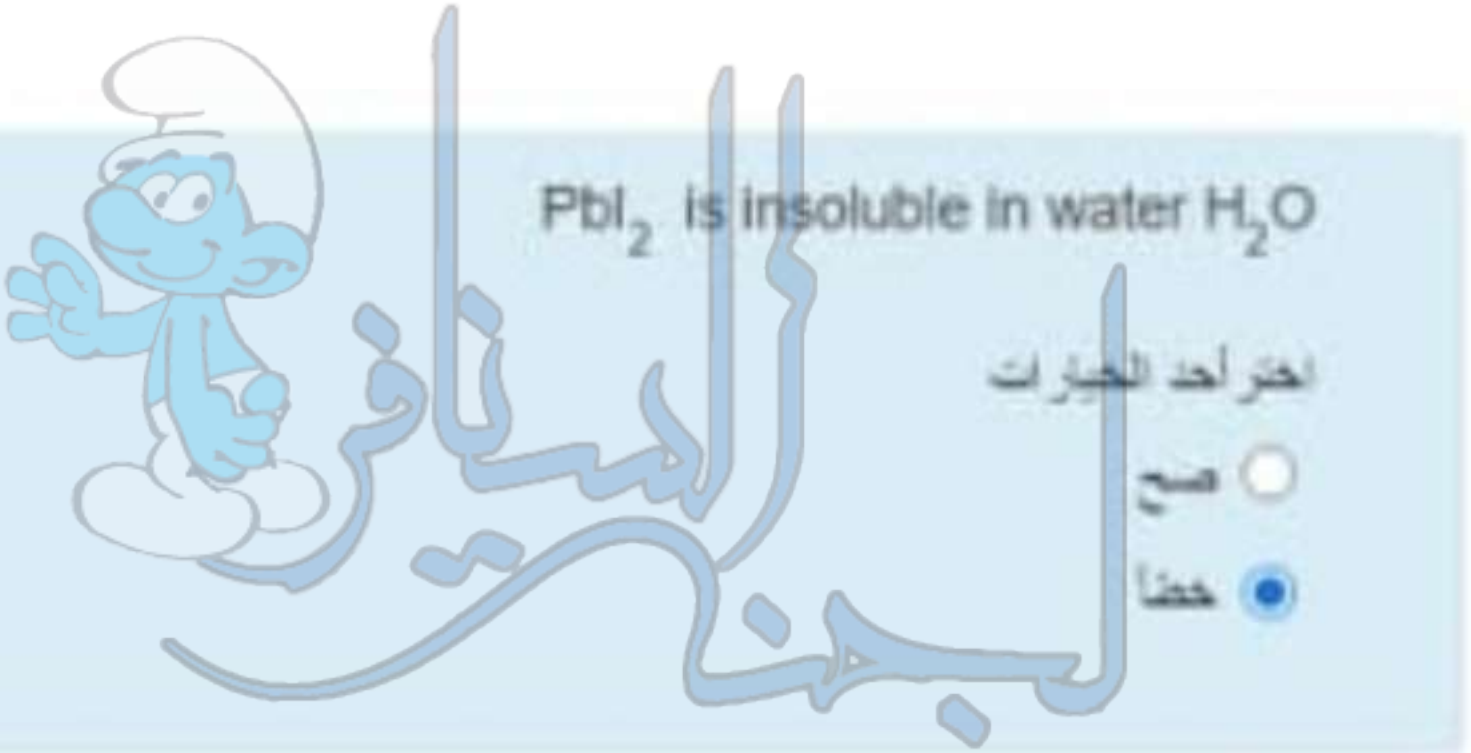
C. not effected



PbI_2 is insoluble in water H_2O

اختر احد الخيارات

- صح
- خطأ



:Which of the following pressure will give you the highest boiling point of Acetone

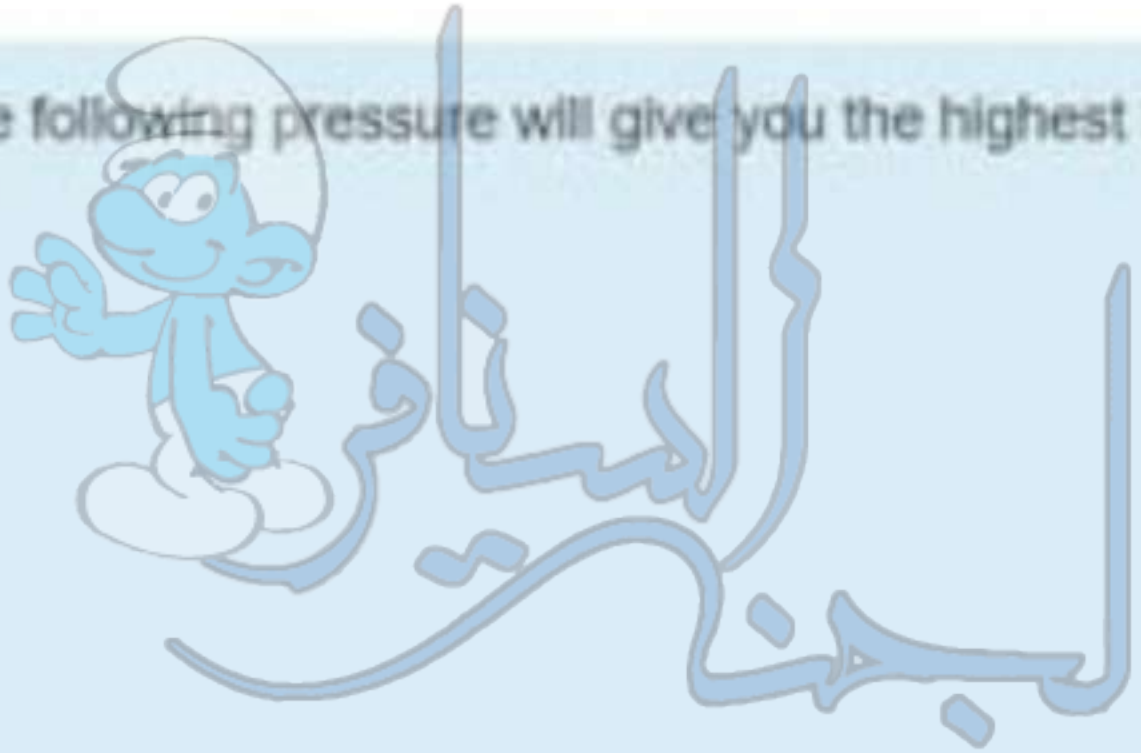
اختر أحد الخيارات

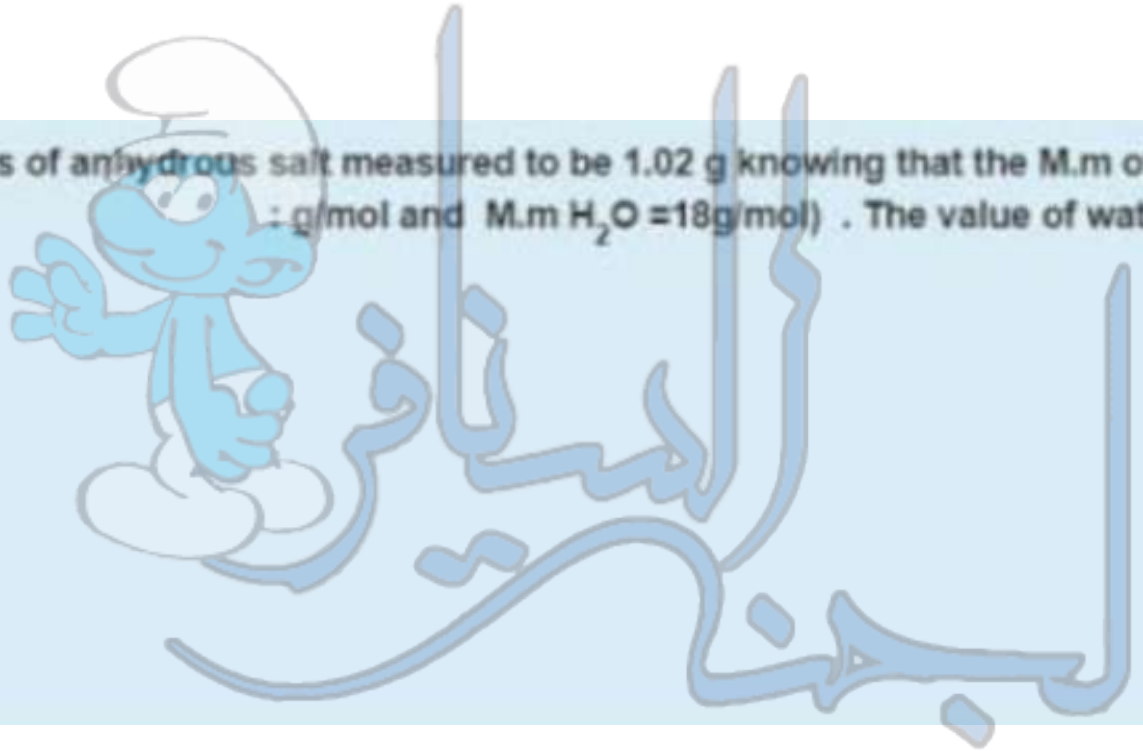
A. 0.52 atm

B. 1 atm

C. 1.85 atm

D. 1.5 atm





salt is heated then the mass of anhydrous salt measured to be 1.02 g knowing that the M.m of anhydrous salt equal = 150 : g/mol and M.m H_2O =18g/mol) . The value of water of crystallization (X) equal

اختر أحد الخيارات

A. 16

B. 8

C. 12

D. 10

??During density determination experiment a student collect the following set of data for density of solid: Then the density of the solid equal

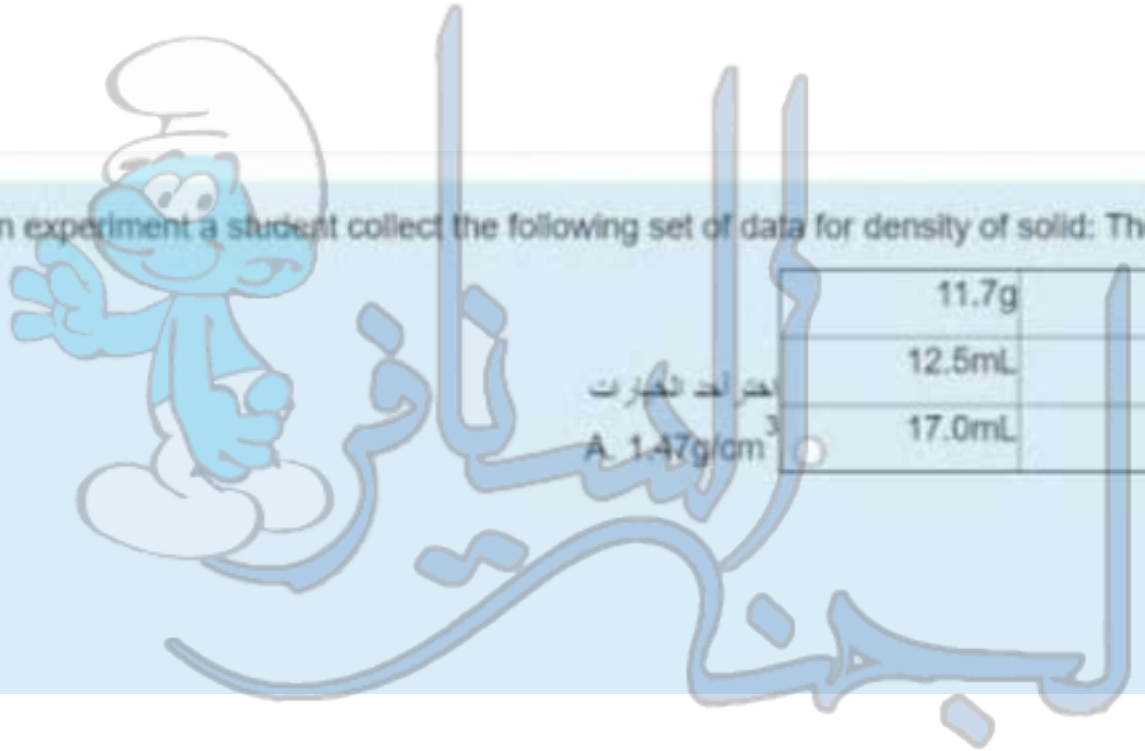
Mass of solid	11.7g
Initial volume of water	12.5mL
Final volume of water	17.0mL

A. 1.47g/cm^3

B. 2.60 g/ cm^3

C. 1.09 g/ cm^3

D. 1.23 g/ cm^3



: Measuring the boiling point of a compound can be used to



اختر احد الخيارات

A. Determine the compound's structure

B. Calculate the compound's density

C. Confirm the compound's identity

D. Determine the compound's vapor pressure



This pictograms indicates that the chemical compound is:

اختر احد الخيارات

A. Toxic

B. Explosive

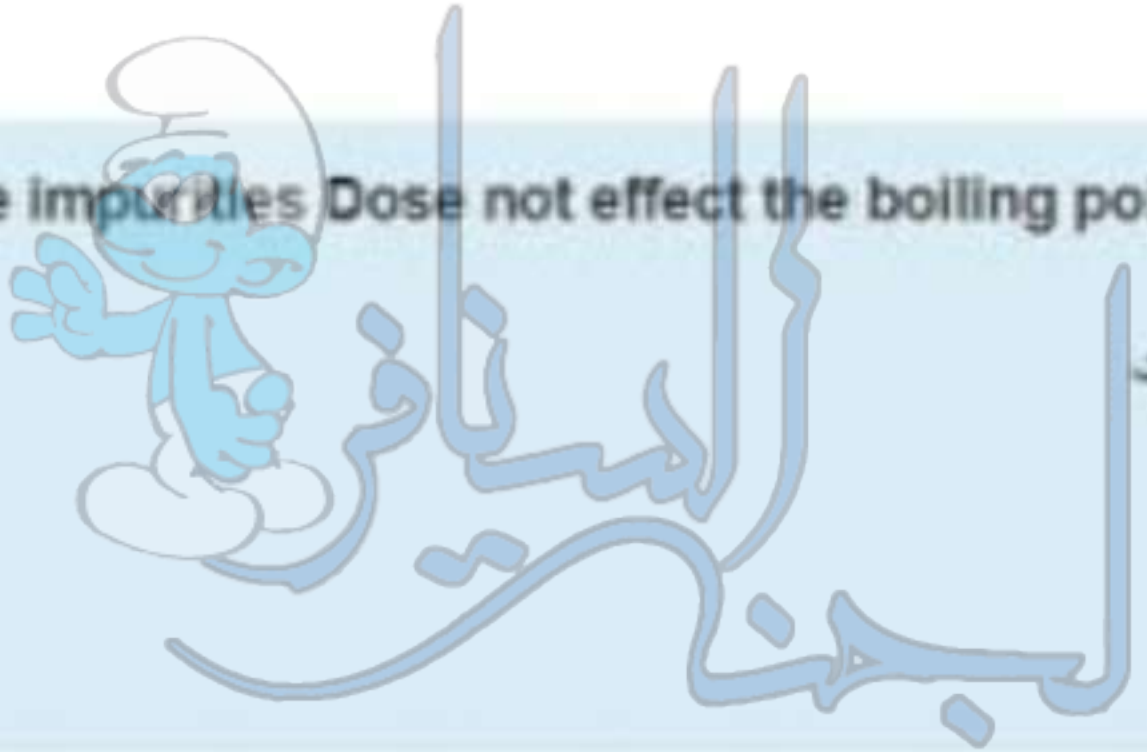
C. flammable

D. Corrosive

Insoluble impurities Dose not effect the boiling point of liquid

اختر احد الخيارات

- صح
- خطا



Which equipment is used to measure the mass of a liquid in density experiment?



اخترأحد الخيارات

A. Test tube

B. Balance

C. Thermometer

D. Capillary tube

The incorrect statement of the following is:

Select one:

- A. To avoid chemical loss, unused chemicals should be returned to the stock bottles.
- B. When dealing with flammable liquid do not heat in blue flame directly.
- C. Always discard waste according to your instructor order.
- D. You can't carry reagent bottles to your desk and take a sample.

Which one of the following will dissolve in Hexane ?



اخترأحد الخيارات

A. Br_2

B. H_2O

C. NaCl

D. CH_3OH

Which one of the following will dissolve in Hexane ?



اخترأحد الخيارات

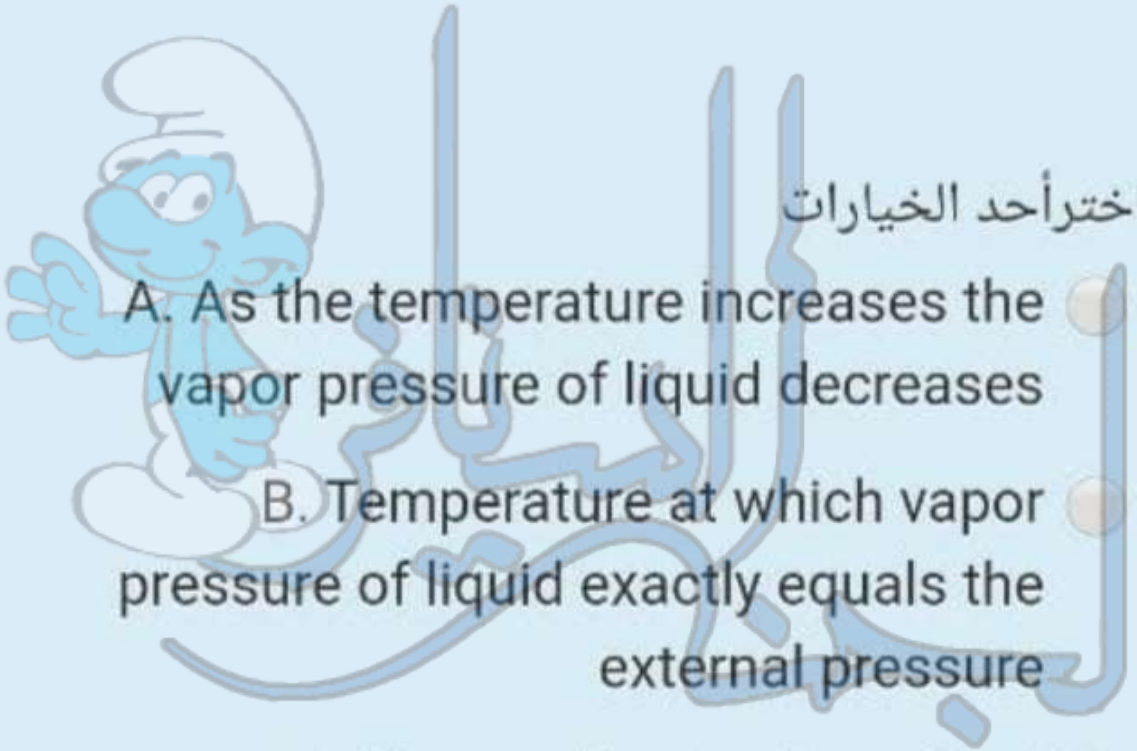
A. Br_2

B. H_2O

C. NaCl

D. CH_3OH

All the following is true regarding the boiling point of liquid except?



اخترأحد الخيارات

A. As the temperature increases the vapor pressure of liquid decreases

B. Temperature at which vapor pressure of liquid exactly equals the external pressure

C. It is used to check purity of compound

D. As the external pressure increases the boiling point increases

Which one of the following will dissolve in Hexane ?



اختر أحد الخيارات

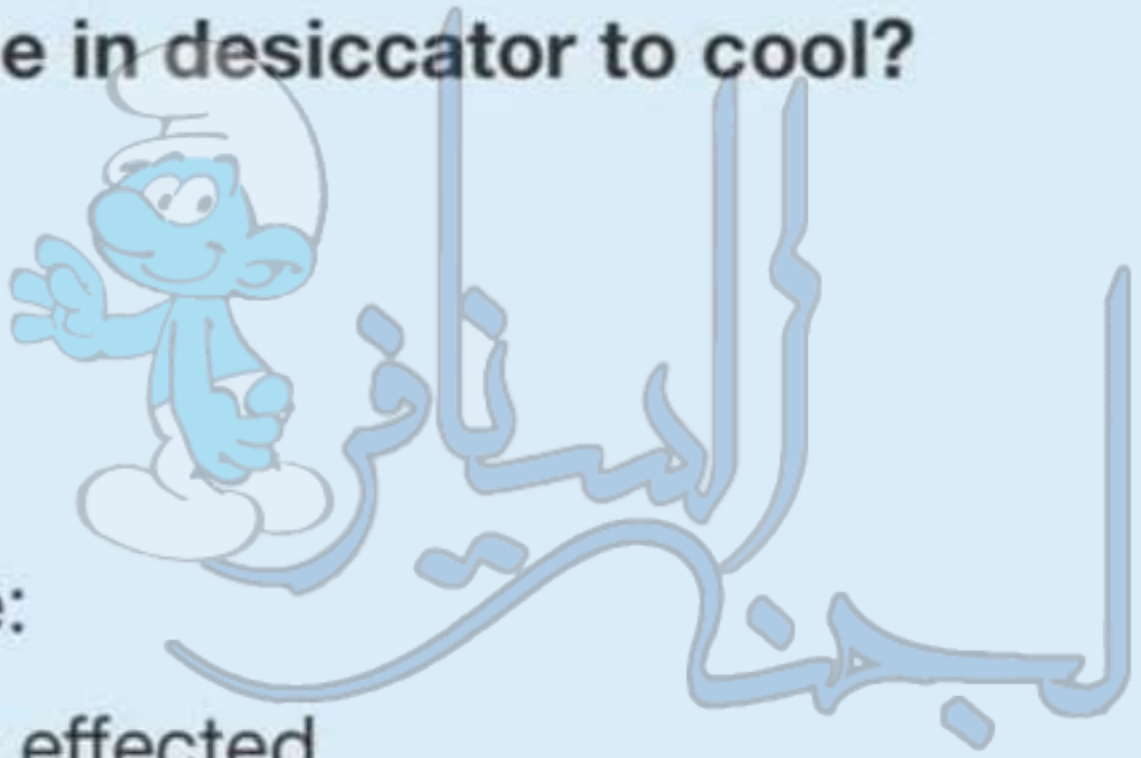
A. Br_2

B. H_2O

C. NaCl

D. CH_3OH

During the hydrated salt Experiment the the value of X will be ----------if we don't put the sample in desiccator to cool?



Select one:

- A. not effected
- B. Decrease
- C. Increase

Which one of the following will dissolve in ?Hexane (C_6H_{14})



اخترأحد الخيارات

A. I_2

B. NaCl

C. CH_3OH

D. H_2O

During density determination experiment a student collect the following set of data for density of solid: Then the density of the solid equal ??

Mass of solid	7.5g
Initial volume of water	12.5mL
Final volume of water	17.6mL

Select one:

- A. 1.09 g/ cm^3
- B. 1.23 g/ cm^3
- C. 1.47g/cm^3
- D. 2.60 g/ cm^3

A 1.603 g of calcium sulfate hydrate salt , ($CaSO_4.XH_2o$), is heated in a crucible until a constant mass is reached. If the mass of of anhydrous $CaSO_4$ salt equal 1.126g.Calculate the percent by mass of water in the hydrated calcium sulfate.

(M.mass : H_2O=18g/mol ,$CaSO_4$ =136.1g/mol)

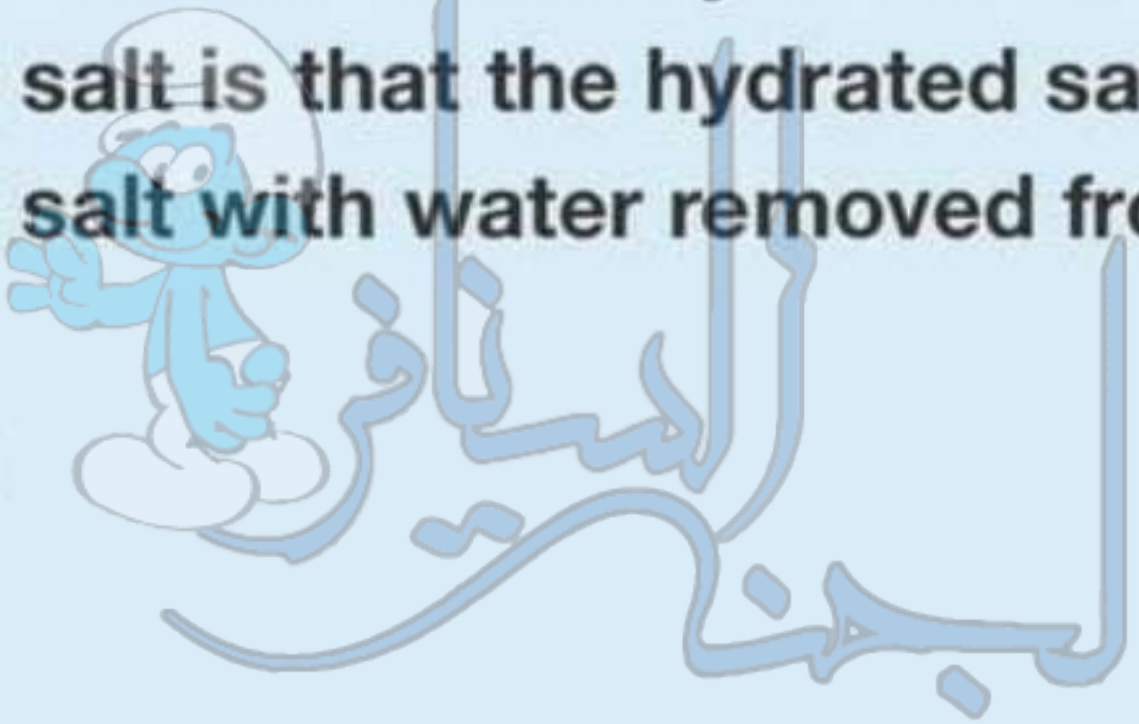
Select one:

- A. 0.79 %
- B. 29.8 %
- C. 2.09 %
- D. 79.0 %

The difference between a hydrated salt and an anhydrous salt is that the hydrated salt is an anhydrous salt with water removed from.

Select one:

- True
- False



Which one of the following will dissolve in ?Hexane (C_6H_{14})



اخترأحد الخيارات

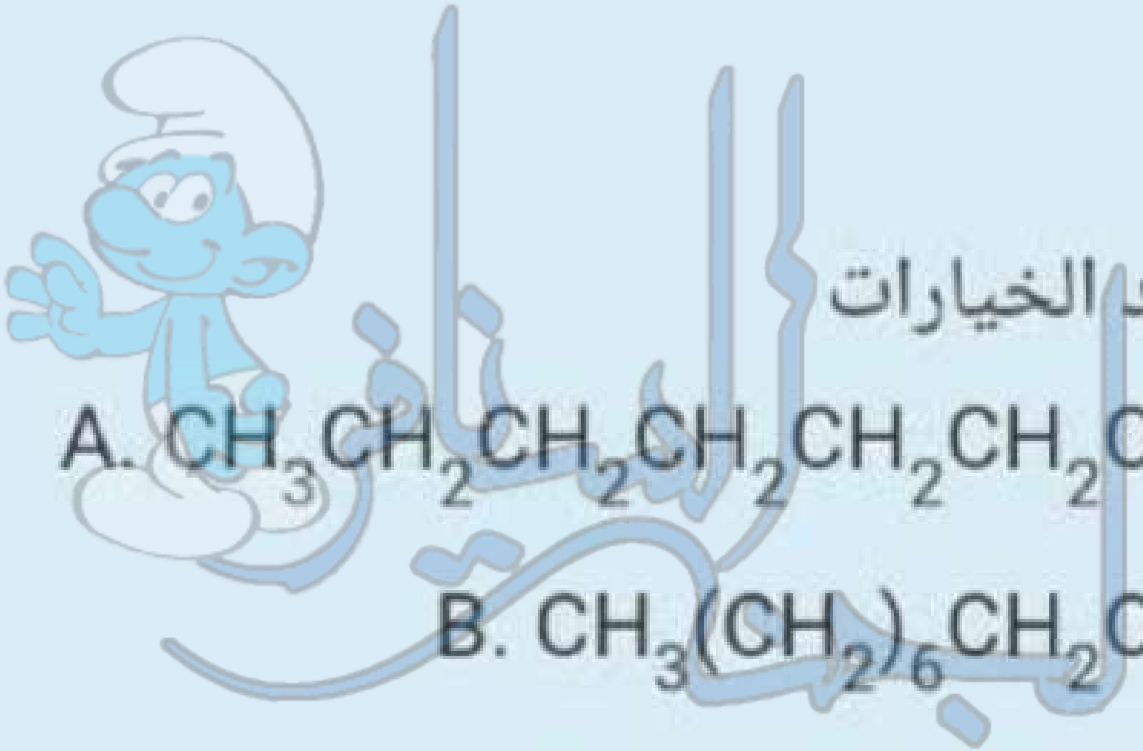
A. I_2

B. NaCl

C. CH_3OH

D. H_2O

Which of the following compounds have the Lower boiling point?



اختر أحد الخيارات

A. $CH_3CH_2CH_2CH_2CH_2CH_2CH_2OH$

B. $CH_3(CH_2)_6CH_2OH$

C. $CH_3CH_2CH_2OH$

D. $CH_3(CH_2)_7CH_2OH$

:The solubility is defined as

اخترأحد الخيارات

A. The amount of material that dissolves in a given amount of solution.

B. The amount of material that dissolves in a given amount of solvent.

C. The amount of solvent that dissolves in a given amount of solute.

D. The amount of material that dissolves in a given amount of solid.

Which one of the following will dissolve in Hexane C_6H_{14} ?

Select one:

- A. C_6H_{12}
- B. NaCl
- C. H_2O
- D. KBr



Which one of these salts is an Alum??

Select one:

- A. $CaMg(SO_4)_2.10H_2O$
- B. $CuSO_4.5H_2O$
- C. $KAl(SO_4)_2.12H_2O$
- D. $NaAl(SO_3)_2.12H_2O$

A 1.503 g of calcium sulfate hydrate salt ,($CaSO_4.XH_2o$), is heated in a crucible until a constant mass is reached. If the mass of of anhydrous $CaSO_4$ salt equal 1.126g.Calculate the percent by mass of water in the hydrated calcium sulfate.

(M.mass : H_2O=18g/mol ,$CaSO_4$ =136.1g/mol)

Select one:

- A. 75.0 %
- B. 0.75 %
- C. 2.19 %
- D. 25.1 %

Which of the following pressure will give you the lowest boiling point of ethanol ??

Select one:

- A. 0.52 atm
- B. 1.5 atm
- C. 0.95 atm
- D. 1 atm



Which one of the following will dissolve in Hexane(C_6H_{14}) ?

Select one:

- A. KBr
- B. H_2O
- C. C_6H_6
- D. NaCl



What is the name of the following



glassware?

Select one:

- 1. Funnel
- 2. Volumetric Flask
- 3. pipette
- 4. burette

Which of the following will Decrease the molar ??mass of volatile liquid if its value increase

إختر أحد الخيارات:

A. Mass of gas m

B. Temperature T

C. Volume of gas V

D. The gas constant value R

Which of the following will not effect the molar mass of volatile liquid??

Select one:

- A. Pressure of gas P
- B. The gas constant value R
- C. Volume of gas V
- D. Temperature T

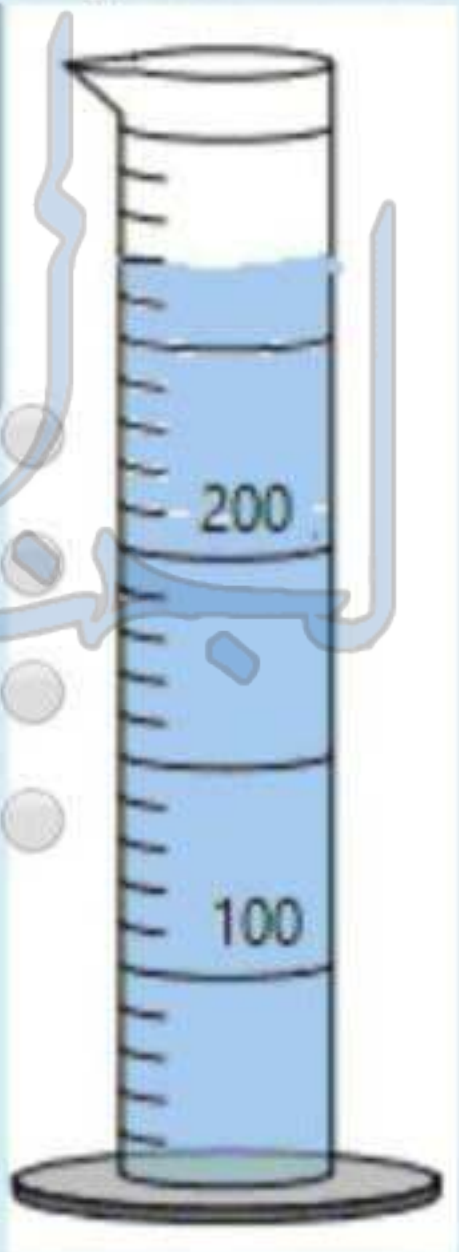


This pictograms indicates that the chemical compound is:

Select one:

- A. Corrosive
- B. Biohazard
- C. harmful
- D. flammable

Calculate the molar mass of an unknown gas sample occupied a 250 mL flask that has a mass of 0.50 g at temperature 25 °C and pressure 1.04 atm. R= 0.0821 atm.L/mol.K, the volume of water occupies the flask shown in the cylinder below:



إختر أحد الخيارات:

A. 47.0 g/mol

B. 40.12 g/mol

C. 43.6 g/mol

D. 51.7 g/mol

A density of a liquid equal 1.82 g/L was evaporated if the temperature was 353.5 K at 798 mmHg then its molar mass is: (R=0.0821 atm.L/mol.K)

Select one:

- A. 60 g/mol
- B. 42.3 g/mol
- C. 85 g/mol
- D. 50.3 g/mol

Consider the set up on the right which used during vinegar analysis. The names of equipment 1 and 2 were ..... & .....respectively. In equipment 1, a solution of ..... was introduced to titrate ------- as the active ingredient in vinegar.

Select one:

- A. Burette, flask, NaOH,CH3COOH
- B. Flask, Burette, NaOH, CH3COOH
- C. Flask, Burette, CH3COOH,NaOH
- D. Burette, flask, NaOH, CH3COOH

Which equipment is used to measure the mass of a liquid in density experiment?

Select one:

- A. Thermometer
- B. Capillary tube
- C. Balance
- D. Test tube

Which one of these salts is an Alum:

Select one:

- A. $KAl(SO_3)_2 \cdot 12H_2O$
- B. $CaMg(SO_4)_2 \cdot 10H_2O$
- C. $NaAl(SO_4)_2 \cdot 12H_2O$
- D. $CuSO_4 \cdot 5H_2O$

If the % mass of H_2O = 37.5%, and the molar mass (Mm) of anhydrous salt = 379 g/mol. The value of water of crystallization (X) .(is: (Mm H_2O =18g/mol

إختر أحد الخيارات:

A. 12

B. 16

C. 13

D. 10



Which equipment is used to measure the volume of a solid in density determination experiment?

Select one:

- A. Graduated cylinder
- B. Thermometer
- C. Volumetric cylinder
- D. Balance

Which equipment is used to measure the mass of a liquid in density experiment?

Select one:

- A. Balance
- B. Thermometer
- C. Capillary tube
- D. Test tube



This pictograms indicates that the chemical compound is:

Select one:

- A. flammable
- B. Toxic
- C. Oxidizing
- D. Corrosive



This pictograms indicates that the chemical compound is:

Select one:

- A. flammable
- B. Explosive
- C. Corrosive
- D. Toxic

The incorrect statement of the following is:

Select one:

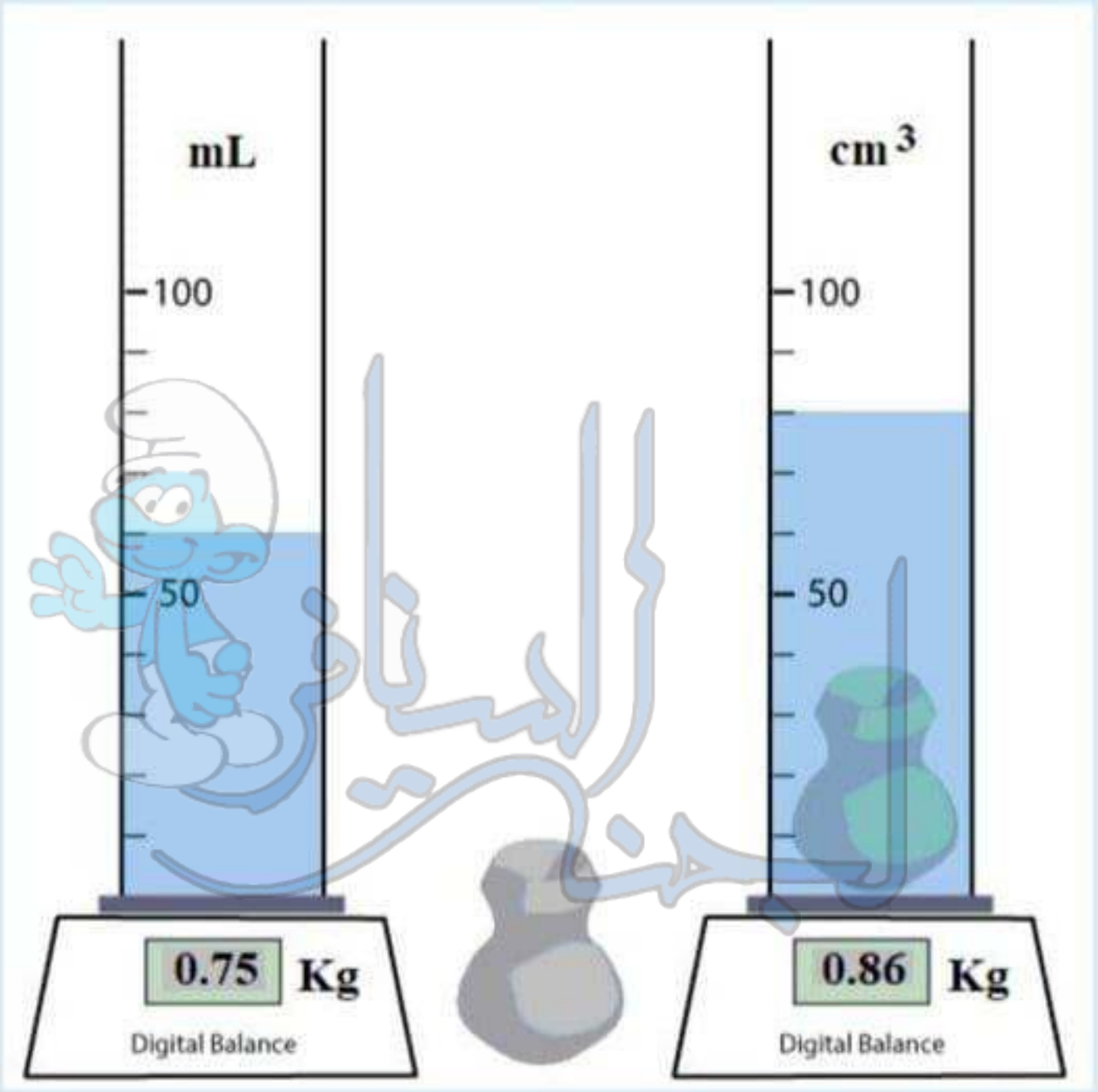
- A. Always prepare your experiment before take it in Lab.
- B. You can carry reagent bottles to your desk and take a sample.
- C. When dealing with flammable liquid heat in water bath.
- D. Unused chemicals should not be returned to the stock bottles.

Which of the following pressure will give you the lowest boiling point of Acetone:

Select one:

- A. 0.95 atm
- B. 1 atm
- C. 0.52 atm
- D. 1.5 atm

The Density of material equal _____
according the photo beside:



Select one:

- A. 0.055 g/mL
- B. 55×10^{-3} g/mL
- C. 5.5×10^{-3} g/mL
- D. 5.5 g/mL

Which of the following will dissolve in water to produce a clear solution?

Select one:

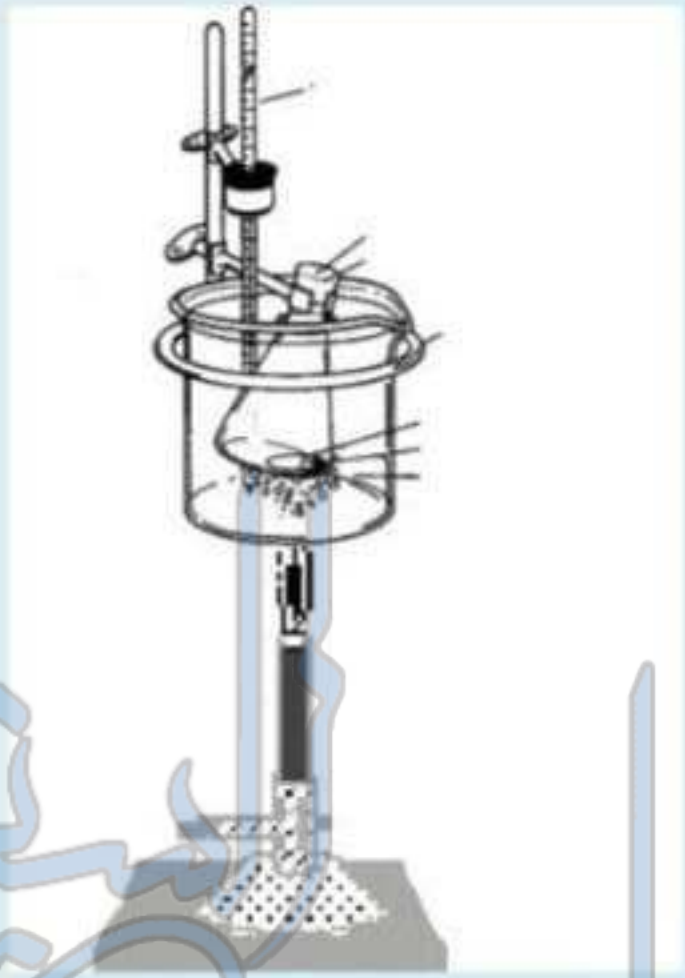
- A. $MgCl_2$
- B. $pbBr_2$
- C. $CH_3(CH_2)_8OH$
- D. AgBr

Calculate the molecular weight of 1.85g of a volatile liquid (condensed vapor) which occupied a volume of 0.685 L at 0.932 atm and 379K (R= 0.0821 atm.L/K.mol):

Select one:

- A. 112.3g/mol
- B. 90.2 g/mol
- C. 83.7 g/mol
- D. 95.6 g/mol

In which experiment dose this apparatus used ?



إختر أحد الخيارات:

A. Empirical formula of ionic compound

B. Water of crystallization

C. Boiling point identification

D. Molecular weight of volatile liquid

During density determination experiment a student collect the following set of data for density of solid: Then the density of the solid equal??

Mass of solid	11.7g
Initial volume of water	12.5mL
Final volume of water	17.0mL

Select one:

- A. $1.47 g/cm^3$
- B. $2.60\ g/\ cm^3$
- C. $1.09\ g/\ cm^3$
- D. $1.23\ g/\ cm^3$

The incorrect statement of the following is:

Select one:

- A. A 10mL graduated cylinder is more precise than a 10mL pipet.
- B. Fume hood is the suitable equipment to resist toxic vapor.
- C. Lab coat should be worn all the lab time.
- D. You have to tell your instructor when the mercury thermometer is broken.

Which of the following will show one layer solution with water (H_2O):

Select one:

- A. C_6H_{14}
- B. Olive Oil
- C. C_6H_6
- D. H_2SO_4



This pictograms indicates that the chemical compound is:

Select one:

- A. Explosive
- B. flammable
- C. Toxic
- D. Corrosive

If the % mass of H_2O = 27.5%, and the molar mass (Mm) of anhydrous salt = 279 g/mol. The value of water of crystallization (X) is: (Mm H_2O =18g/mol).

Select one:

- A. 13
- B. 6
- C. 16
- D. 9

Which of the following will increase the molar mass of volatile liquid if its increase??

Select one:

- A. The gas constant value R
- B. Volume of gas V
- C. Mass of gas m
- D. Pressure of gas p